MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1906.3
STANDARD	
Motion Imagery Metadata (MIMD): Staging System	**2025-06-11**

1 Scope

The Motion Imagery Metadata (MIMD) Model [1] includes locations, orientations, and kinematics (velocity, acceleration, etc.) of platforms, gimbals, sensors, sensor elements, geospatial and other points. The locations and orientations are either absolute references to a well-known frame of reference (e.g., WGS84 [2]) or relative references to other locations and orientations. Each location and orientation pairing define a "stage" that has the potential to be the frame of reference for another location and orientation. Linking stages together forms the Staging System. The Staging System then defines an ability to describe, in metadata, the physical make-up and configuration of a system and the time varying physical relationships of the system and its sub-system components.

2 References

[1] MISB, *ST 1903.3 - Motion Imagery Metadata (MIMD): Model,* 2025-06-11.

[2] WGSG, *NGA.STND.0036_1.0.0 - DoD World Geodetic System 1984,* 2014-07-08.

[3] Jack Kuipers, Quaternions and Rotation Sequences: A Primer with Applications to Orbits, Aerospace, and Virtual Reality, Princeton, New Jersey: Princeton University Press, 1999.

[4] MISB, *ST 1504.1 - Natural Representation of Orbital State Vectors,* 2018-11-01.

[5] MISB, *ST 1901.3 - Motion Imagery Metadata (MIMD): Modeling Rules,* 2025-06-11.

[6] MISB, *ST 1907.3 - Motion Imagery Metadata (MIMD): Payload,* 2025-06-11.

[7] MISB, *MISP-2025.1: Motion Imagery Handbook,* 2025-06-11.

[8] MISB, *ST 1904.3 - Motion Imagery Metadata (MIMD): Base Attributes,* 2025-06-11.

[9] ASM, *NGA.STND.0033_3.0.1 - Geopolitical Entities, Names, and Codes (GENC) Standard,* 2016-07-08.

3 Revision History

Version	Approved Date	Change Description
ST 1906.3	2025-06-11	• Updated notation ○ Use Position (P) and Rotations (Q) instead of "S" based notation.

		○ Use "above" text, i.e., $\overset{ENU}{Q}$ to clearly state purpose ○ Added notational summary Section 4.3 • Reorganized and added clarifications to some sections. • Added Absolute Geocentric orientation class (AbsECEF) • Allow for non-linear staging configurations ○ Added section on stage configurations ○ Added reverse methods section ○ Retired requirement -04 (in ST 1907 added gscpoStage attribute) • Updated country code to support ISO 3166 ○ Added requirement for all U.S. systems to use GENC only ○ Added a requirement to follow GENC short URN format • Removed restriction for some classes to contain all three values to be valid. ○ Retired requirements, -03, -06, -08 & -09 • General Editing

4 Definitions of Terms, Acronyms, and Initialisms

4.1 Terms

The following lists the terms and their definitions this document uses.

Constellation	A collection of one or more stages
Root Stage	A stage which is the starting position and orientation for child stages; does not have a parent
Stage	A single frame-of-reference point with information about its location, orientation, and kinematics
Terminal Stage	A stage that does not have a child; the end stage in a chain of stages starting at the root and ending at the terminal stage

4.2 Acronyms and Initialisms

The following lists the acronyms and initialisms this document uses.

CG	Center of Gravity
CP	Center Point
ECEF	Earth-centered earth-fixed
ENU	East-North-Up
GSCPO	Geo-Intelligence Sensor Common Pointing Orientation (see Section 6.5)
HAE	Height above Ellipsoid
IERS	International Earth Rotation and Reference Systems Service
IRM	IERS Reference Meridian
MIMD	Motion Imagery Metadata
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
NGA	National Geospatial-Intelligence Agency
ROC_Unknown	Report-on-Change Unknown state (see MISB ST 1901)
SI	International System of Units

ST	Standard
UML	Unified Modeling Language
WGS84	World Geodetic System 1984

4.3 Notation Summary

The following lists the notation this document uses.

Notation	Description
$S_i = (P_i, \mathrm{Q_i})$	The i[th] stage contains a staging position (P_i) and staging orientation (Q_i).
$P_i = \begin{cases} {}^{\mathrm{LLH}}P_i\ (\phi, \lambda, h) \\ {}^{\mathrm{XYZ}}P_i\ (x, y, z) \\ \vec{P}_i(x, y, z) \\ \overset{\angle}{P}(\Psi, \Omega, M) \end{cases}$	Staging position is either: • Geodetic = ${}^{\mathrm{LLH}}P_i$ • Geocentric = ${}^{\mathrm{XYZ}}P_i$ • Relative = $\vec{P}_i$ • Relative-Angular = $\overset{\angle}{P}$
${}^{\mathrm{LLH}}P_i\ (\phi, \lambda, h)$	Absolute WGS84 Geodetic Position with latitude (ϕ), longitude (λ), and height above ellipsoid (h).
${}^{\mathrm{XYZ}}P_i\ (x, y, z)$	Absolute geocentric Position with distance in meters from geocentric earth origin in x, y, and z axis.
$\vec{P}_i(x, y, z)$	Relative cartesian position with distance in meters from parent stages origin along the parent stage's x, y, and z axis.
$\overset{\angle}{P}(\Psi, \Omega, M)$	Relative angular position with azimuth angle (Ψ), elevation angle (Ω), and magnitude (M) relative to the parents' stage.
$Q_i = \begin{cases} {}^{\mathrm{ENU}}Q_i\ (\alpha, \beta, \psi) \\ {}^{\mathrm{ECEF}}Q_i\ (\alpha, \beta, \psi) \\ \widetilde{Q}_i(\alpha, \beta, \psi) \end{cases}$	Staging orientation is either: • Absolute to ENU = ${}^{\mathrm{LLH}}Q_i\ (\alpha, \beta, \psi)$ • Absolute to ECEF = ${}^{\mathrm{ECEF}}Q_i\ (\alpha, \beta, \psi)$ • Relative = $\widetilde{Q}_i(\alpha, \beta, \psi)$
${}^{\mathrm{ENU}}Q_i\ (\alpha, \beta, \psi)$	Absolute ENU orientation is the angular rotations (α, β, ψ) around the ENU-LRP. The rotations are around the Z-axis (α), new-Y-axis (β), and new-new-X-axis (ψ).
${}^{\mathrm{ECEF}}Q_i\ (\alpha, \beta, \psi)$	Absolute ECEF orientation is the angular rotations (α, β, ψ) around the earth's ECEF axis. The rotations are around the Z-axis (α), new-Y-axis (β), and new-new-X-axis (ψ).
$\widetilde{Q}_i(\alpha, \beta, \psi)$	Relative orientation is the angular rotations around a parent stage's coordinate system. The rotations are around the Z-axis (α), new-Y-axis (β), and new-new-X-axis (ψ).

5 Introduction

A Motion Imagery System needs to specify position, orientation, and kinematics (velocity, acceleration, etc.) for both physical components and geospatial points of interest. The positions, orientation, and kinematics may be measures to well-known frame of references, for example WGS84 or East-North-Up (ENU), or measures to a local coordinate system defined by the

Motion Imagery System. For example, the position of the Motion Imagery platform may be in WGS84 latitude, longitude, and height above ellipsoid coordinates, while the position of the sensor onboard the platform may be the location X, Y, and Z meters relative to the center of gravity of the platform or some other benchmark on the platform. When position, orientation, and kinematics are measures based on a well-known frame of reference, the measures are *Absolute*; otherwise, they are *Relative*. Often exploiters are interested in computing the absolute position/orientation for a sensor with a position/orientation relative to an absolute position/orientation. By combining the absolute position/orientation and all the relative positions of preceding components the absolute position/orientation can be determined, as illustrated by the following example.

Figure 1 shows an image of a four-camera ground-based tracking system and a representation of the linkage of the components. Starting from the lower right of the diagram, the *Reference Frame Survey Point* defines the absolute position and orientation of the system relative to WGS84 (or ECEF) and East-North-Up (ENU). The *Trailer* defines a Center Point (CP) and it has a relative position, or translation in X, Y, and Z to the *Reference Frame Survey Point* along with an orientation relative to the *Reference Frame Survey Point's* orientation. The locations of the *Left Pan & Tilt* and *Right Pan & Tilt* cameras are relative translations and orientations from the CP and operate on their own gimbals. The locations and orientations of the *Upper Left* and *Upper Right* cameras are relative to the *Top Center Gimbal* which itself is a relative translation and orientation to the CP.

This example shows the interconnectivity of the different components and the impact each component can have on other components. For example, with this system of cameras a rotation of the trailer affects the absolute positioning and orientation of each sensor because each camera mount is dependent on the position and orientation of the preceding component. The method of communicating all the points, their configuration and movement is a primary capability provided by the Staging System.

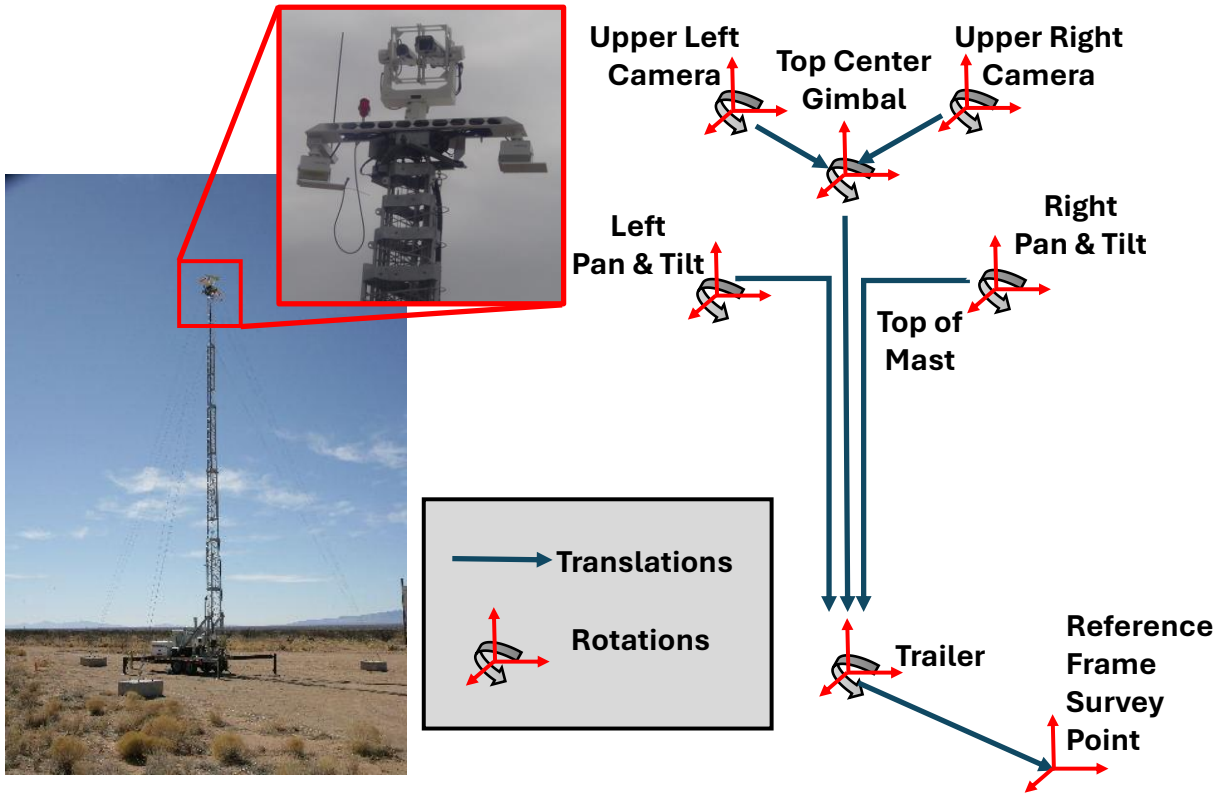


Figure 1: Example Motion Imagery System with Multiple Sensors and Gimbals (Photo credit White Sands Missile Range)

6 Staging System Overview

A *stage* is a single frame-of-reference point with information about its location, orientation, and kinematics. For example, in Figure 1 each red axis ( ) is a stage with a position (either absolute or relative), orientation (absolute or relative), and kinematics.

A *constellation* is a collection of one or more stages. For example, Figure 1 is a constellation.

The Staging System defines the rules for describing and organizing the overall positions, orientations, and kinematics of all stages within, or viewed by, a Motion Imagery System. The Staging System is based on a hierarchy of stages with each stage having a parent and zero or more children. A *root* stage provides the starting position and orientation and does not have a parent. Each child stage measures its position, orientation, and kinematics relative to its parent. For example, in Figure 1 the *Reference Frame Survey Point* is the root stage or starting point of the constellation. The Trailer measurement relative to the root is the first child of the Staging System. The position is the distance in X, Y, Z to the root and the orientation are the angles α, β, ψ (rotation about z, y, and x) relative to the root axis. The *Left Camera* measurement, relative to the Trailer measurement, is the second child of the Staging System. Like the first child's measurements, the second child's position and orientation are relative positions to the first child. Likewise, all remaining child measurements are relative to their physical parents' position and orientation.

Table 1 lists the constellation for Figure 1 as a hierarchical organization, or outline, of the stages. Stage 1 is the root stage and in this example; it uses an absolute frame-of-reference for its measurements (in red). Stage 1 contains a child stage, Stage 2, that has a relative position and orientation (in blue), relative to Stage 1. Stage 2 also has its child stages, Stage 3, 4, and 5, with each position and orientation relative to Stage 2. This hierarchical pattern continues for all stages.

Table 1: Example Constellation Hierarchy

Stage 1 - Reference Frame Survey Point
Absolute Position/Orientation
Stage 2 – Trailer
Relative Position / Orientation
Stage 3 – Left Camera
Relative Position / Orientation
Stage 4 – Right Camera
Relative Position / Orientation
Stage 5 – Top Center Gimbal
Relative Position / Orientation
Stage 6 – Left Upper Camera
Relative Position / Orientation
Stage 7 – Right Upper Camera
Relative Position / Orientation

6.1 Staging System Definitions/Notations

Each stage includes metadata for a position and an orientation, which are either absolute or relative. Absolute means the measurements are in the context of a well-known datum, such as WGS84 or East-North-Up (ENU). Relative means a stage's measurements are in the context of its parent stage.

As a collection of one or more stages a *constellation* always includes a root stage with optional child stages. The root stage defines the frame of reference for the constellation in which all child stages resolve to. For example, if the root stage is absolute and a child stage is relative to the root, with some computation the absolute position of the child stage is determinable; that is, the child stage's position and orientation resolves to the same frame of reference as the parent.

A *terminal* stage is a stage that does not have a child; it is the end stage in a chain of stages starting at the root and ending at the terminal stage. A constellation has only one root but may have many terminal stages. When a constellation has only the root stage, the root stage is also a terminal stage. Referring to Table 1, Stages 3, 4, 6, and 7 are terminal stages.

The notation for a single stage is S_i, where i is the ith stage in the constellation. Each stage has components for position (P) and orientation (Q), $S_i = (P_i, Q_i)$ The following sections describe the component's notations.

6.1.1 Position Definition

The staging system supports two types of positions, Absolute and Relative, which combine easily.

6.1.1.1 Absolute Position

The Staging System's absolute positions use either WGS84 geodetic or geocentric coordinates.

6.1.1.1.1 Absolute Position with Geodetic Coordinates

WGS84 geodetic coordinates use the geodetic Latitude, Longitude, and Height above Ellipsoid (HAE) to define a position below, on, or above the WGS84 ellipsoid. This standard represents a stage's absolute geodetic position using "LLH" (i.e., Latitude, Longitude, HAE) over the letter P, i.e., $\overset{LLH}{P}_i(\phi, \lambda, h)$, where P_i represents the i^{th}stage's position with $\phi, \lambda,$ and h equal to the latitude, longitude, and HAE, respectively. For example, the geodetic position of the Washington Monument is 38°53'22.1"N, 77°02'07.0"W, 12 meters HAE or in decimal 38.889484, -77.035279, 12, which this standard represents as:

$$\overset{\text{LLH}}{P}_i(\phi, \lambda, h) = \overset{\text{LLH}}{P}_i(38.889484, -77.035279, 12)$$

Every geodetic position is the origin of an ENU local reference coordinate system which provides the basis for staging orientation, see Section 6.1.2 for further information.

Systems can transform geodetic coordinates to geocentric coordinates using the algorithms from Appendix B.1.

6.1.1.1.2 Absolute Position with Geocentric Coordinates

WGS84 geocentric coordinates use the Earth-Centered Earth-Fixed (ECEF) Cartesian coordinates (i.e., X, Y, Z) to define a position in three-dimensional space around the origin of the earth. This standard represents a stage's absolute geocentric position using "XYZ" over the letter P, i.e., $\overset{XYZ}{P}_i(x, y, z)$ where P_i represents the i^{th}stage with $x, y,$ and z equal to measurements in meters from the center of the earth along the WGS84 defined axis. For example the geocentric ECEF position of the Washington Monument is:

$$\overset{\text{XYZ}}{P}(x, y, z) = \overset{\text{XYZ}}{P}(1115258.0, -4844319.0, 3982775.0)$$

Geocentric positions also define a vector in three-dimensional space from the earth origin to the point; therefore, this document also uses vector notation, e.g., $\overset{\text{XYZ}}{P}_i(x, y, z) = [x, y, z]^T$.

Every geocentric position is the origin of a ECEF local reference coordinate system which provides the basis for staging orientation, see Section 6.1.2 for further information.

Systems can transform geocentric coordinates to geodetic coordinates using the algorithms from Appendix B.2.

6.1.1.2 Relative Position

The Staging System's relative positions use either Cartesian coordinates (i.e., x, y, z distance measurements in meters) or relative angular vector coordinates (i.e., azimuth angle, elevation angle, and a range value in meters) from the parent stage's coordinate system.

The representation of a stage's relative Cartesian coordinates is a single arrow accent, e.g., $\vec{P}_i(x, y, z)$ over the letter P with $x, y,$ and z equal to measurements in meters. Relative Cartesian coordinates are vectors in three-dimensional space, with the vector origin at the parent stages origin. Since relative Cartesian and absolute Cartesian coordinates are both vectors and use the same units, ECEF vectors and relative vectors combine with standard vector operations (e.g., addition and subtraction).

The representation of a stage's relative angular vector is the angle accent, e.g., $\overset{\angle}{P}(\Psi, \Omega, M)$ over the letter P, where Ψ is the azimuth angle, Ω is the elevation angle, and M is the magnitude or range. Relative angular vectors are equivalent to Cartesian coordinates; applications can convert to and from Cartesian coordinates. Equation 1 defines a function, f, which converts a relative angular vector to Cartesian coordinates.

$$\vec{P}(x, y, z) = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = f\left(\overset{\angle}{P}(\Psi, \Omega, M)\right) = M \begin{bmatrix} cos(\Omega)\cos(\Psi) \\ cos(\Omega)\sin(\Psi) \\ sin(\Omega) \end{bmatrix} \qquad \textbf{Equation 1}$$

Figure 2 illustrates three stage's positions, $\overset{\text{XYZ}}{P_1}$, $\overset{\angle}{P_2}$, and $\vec{P}_3$, and their physical relationships. Each stage position defines its own coordinate system, shown with the red (X), green (Y), and blue (Z) axis; the illustration shows each X-Y plane shaded in grey. In this example the orientation of each stage is different than the previous or next stage, Section 6.1.2 describes stage orientation. Stage $\overset{\text{XYZ}}{P_1}$ has an absolute ECEF position. Position $\overset{\angle}{P_2}$ is an angular vector located with the azimuth value of Ψ relative to $\overset{\text{XYZ}}{P_1}$'s X-axis, an elevation angle Ω relative to $\overset{\text{XYZ}}{P_1}$'s X-Y plane, and range of M from the origin of $\overset{\text{XYZ}}{P_1}$. Position $\vec{P}_3$ is a Cartesian position located at point x, y, and z along $\overset{\angle}{P_2}$'s X, Y, and Z axes repectively.

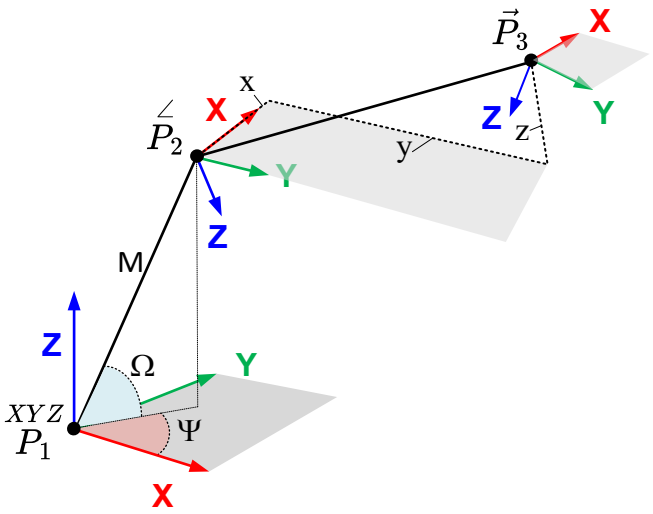


Figure 2: Example of Cartesian and Relative Angle Vector

6.1.1.3 Combining Positions

Combining different systems of absolute position requires converting one or the other to the same reference system using the standard algorithms in Appendix B. Transforming positions between reference systems can introduce small errors which accumulate; thus, an objective is to minimize the number of transformations.

A constellation typically defines an absolute position in the first stage (i.e., the root stage) and one or more child stages as relative stages with measurements in the frame of reference to the root stage. Child stages may be parents to other [grand] children's stages with the [grand] children coordinates' relative to their respective parent.

Figure 3 illustrates a constellation example with a root stage defined by absolute position (shown in blue) and two relative position stages shown in green. The stage 1 position, $\overset{LLH}{P}_1$, has WGS84 Latitude, Longitude, and HAE coordinates, so the constellation's frame of reference is the WGS84 coordinate system. The stage 2 position, $\vec{P}_2$, has relative coordinates to its parent, i.e., 83 meters in the X direction, 33 meters in the Y direction, and zero meters in the Z direction, from the $\overset{LLH}{P}_1$ position. The stage 3 position, $\vec{P}_3$, has relative coordinates to its parent, $\vec{P}_2$.

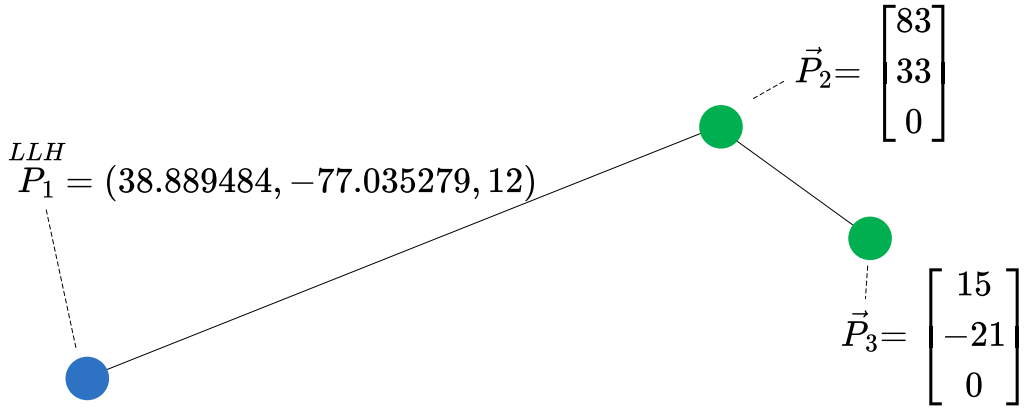


Figure 3: Example of Stage Position with Geodetic Absolute Root Stage

With geodetic absolute position (and ignoring each stage's orientation for this example), the geodetic position of $\vec{P}_3$ can be determined by converting the geodetic coordinate to geocentric (ECEF) coordinates and adding stage values (see Equation 2) then converting the result back to geodetic; the resulting value is equal to (38.88942, -77.03415, 9).

Figure 4 illustrates the same constellation using a geocentric absolute position. The first stage in both examples is the same location on the earth, just different representations.

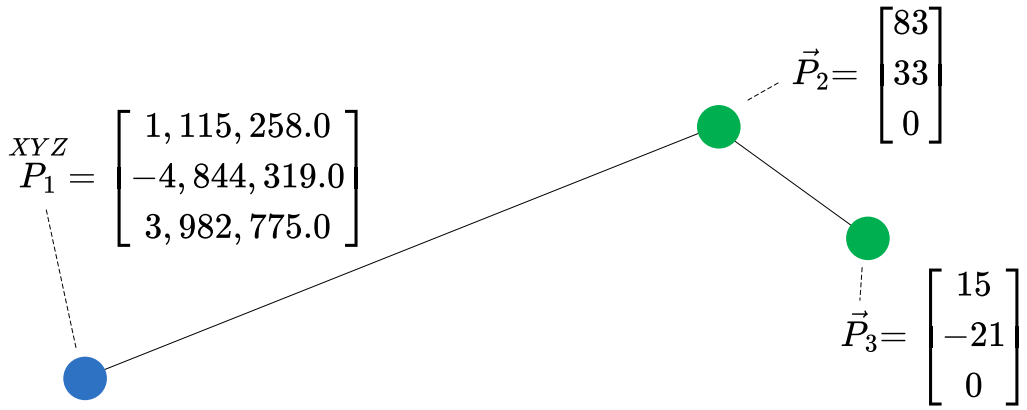


Figure 4: Example of Stage Position with Geocentric Absolute Root Stage

With ECEF absolute position and ignoring each stages orientation, determining the ECEF coordinate of $\vec{P}_2$ and $\vec{P}_3$ uses addition. For example, the coordinates of $\vec{P}_3$ are:

$$^{LLH}P_1 + \vec{P}_2 + \vec{P}_3 = \begin{bmatrix} 1{,}115{,}258.0 \\ -4{,}844{,}319.0 \\ 3{,}982{,}775.0 \end{bmatrix} + \begin{bmatrix} 83 \\ 33 \\ 0 \end{bmatrix} + \begin{bmatrix} 15 \\ -21 \\ 0 \end{bmatrix} = \begin{bmatrix} 1{,}115{,}356.0 \\ -4{,}844{,}307.0 \\ 3{,}982{,}775.0 \end{bmatrix} \quad \text{Equation 2}$$

Figure 5 repeats the example with the last stage using a relative angular vector, i.e., $\angle\vec{P}_3$. The relative angular vector converts to a cartesian vector which operates like P_3 in Equation 2.

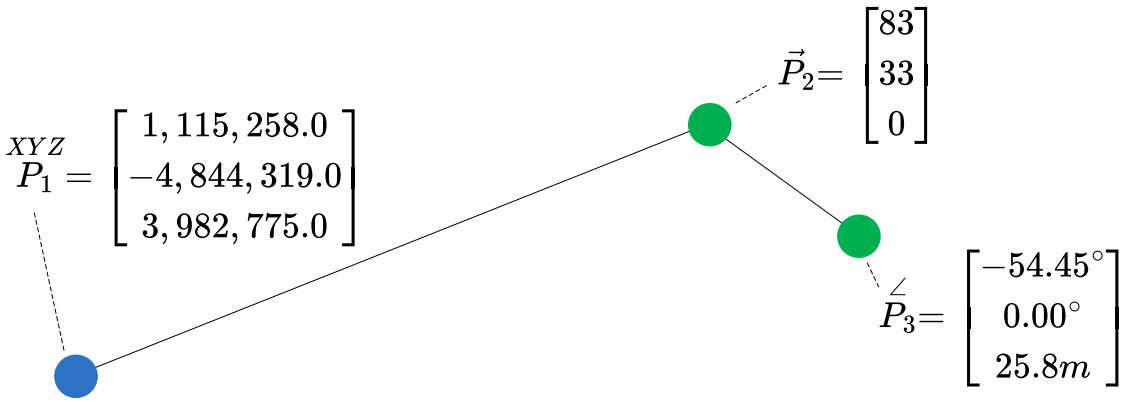


Figure 5: Example of Stage Position with Relative Angular Vector

$$\overset{\angle}{P}_3 = 25.80 \begin{bmatrix} \cos(0)\cos(-54.45) \\ \cos(0)\sin(-54.45) \\ \sin(0) \end{bmatrix} = \begin{bmatrix} 15.00 \\ -21.0 \\ 0 \end{bmatrix}$$ **Equation 3**

If a constellation has a relative root stage, the constellation defines a local coordinate system with no absolute reference. The origin of the constellation is determined from the first stages coordinates. Figure 6 illustrates a constellation example with a relative root stage, $\vec{P}_1$. The stage 1 position is relative to a local coordinate system origin. The origin is located at the negative of the $\vec{P}_1$ coordinates, as shown in grey. This is the origin for the constellation so all points in the constellation resolve to the origin's frame of reference. The stage 2 position is 69 meters in the X direction, 8 meters in the Y direction, and zero meters in the Z direction from $\vec{P}_1$'s coordinates. The stage 3 position is 20 meters in the X direction, -1 meter in the Y direction, and zero meters from $\vec{P}_2$'s coordinates.

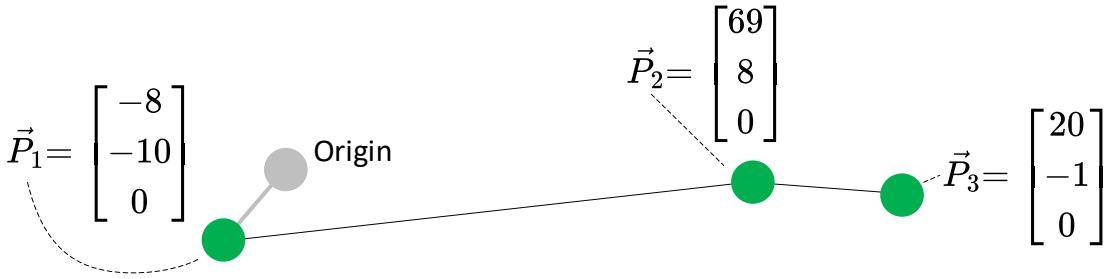


Figure 6: Example of Stage Position in 2-D with Relative Root Stage

6.1.2 Orientation Definition

The Staging System's orientations use Euler rotations, measured in radians, around three axes (X, Y, and Z) of a righthanded coordinate system.

The Staging System standardizes rotations to use a specific order of rotations. The Staging System uses Tait-Bryan method with intrinsic rotations and a Primary Rotation Order of Z-Y-X (see Appendix C); thereby the first rotation is around the Z-axis, the second rotation is around the new Y-axis (after rotation around the Z-axis), the final rotation is around the new-new X-axis (after rotation around the Z and then Y axes) [3].

The staging system supports two types of orientation, Absolute and Relative.

6.1.2.1 Absolute Orientation

Each absolute position type (i.e., Geodetic and Geocentric) defines reference frames relative to the earth, i.e., the orientations of coordinate system's axes relative to the earth's axes. An Absolute Orientation is a set of angles relative to the absolute orientation's axes the stage specifies.

6.1.2.1.1 Absolute Geodetic Orientation

The Staging System's absolute geodetic orientation is a set of rotations around a position-defined coordinate system aligned with the East, North, and Up (ENU). Every absolute position on,

above, or below the WGS84 ellipsoid (except at the center of the earth), defines the origin of a local topographical reference coordinate system. As Figure 7 illustrates, the orientation of the coordinate system has the X-Y plane (yellow) parallel to a plane (blue) which is tangent to the ellipsoid; where the X-axis aligns with the latitude or meridian lines, with positive values going east; the Y-axis aligns with longitude lines, with positive values going north. This standard refers to the X-Y plane (yellow) as the ENU Local Reference Plane (ENU-LRP). The Z-axis is orthogonal to the planes (i.e., geodetically) with positive values increasing in the upwards direction, away from the pole, or WGS84 Z-axis; unless the position is at either pole, then the positive direction is away from the earth's center of gravity.

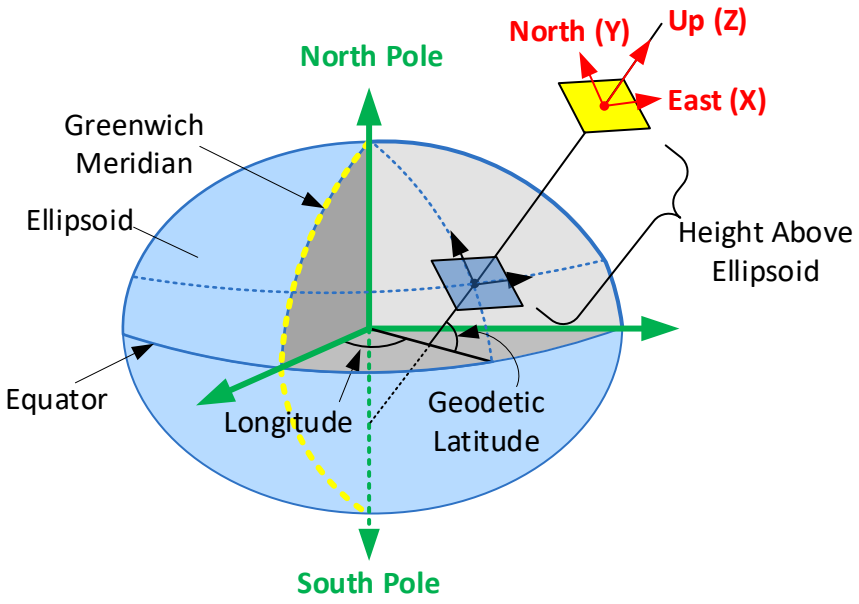


Figure 7: East-North-Up (ENU) Orientation

When the ENU-LRP is orthogonal to the north/south poles (i.e., the ENU-LRP is directly over the north or south pole), the north and east directions are ambiguous and therefore undefined. To assure a fixed reference, this standard defines a specific orientation of the ENU-LRP when the origin of the ENU-LRP is over the north and south poles. Figure 8 illustrates the orientations of the ENU-LRP over either the North Pole or South Pole. In this figure the prime meridian is blue, the anti-meridian is green, and the east/west meridians are orange. On the left of Figure 8, when directly over the North Pole, the North axis of the ENU-LRP aligns to the anti-meridian. On the right side, when directly over the South Pole the North axis of the ENU-LRP aligns with the prime meridian.

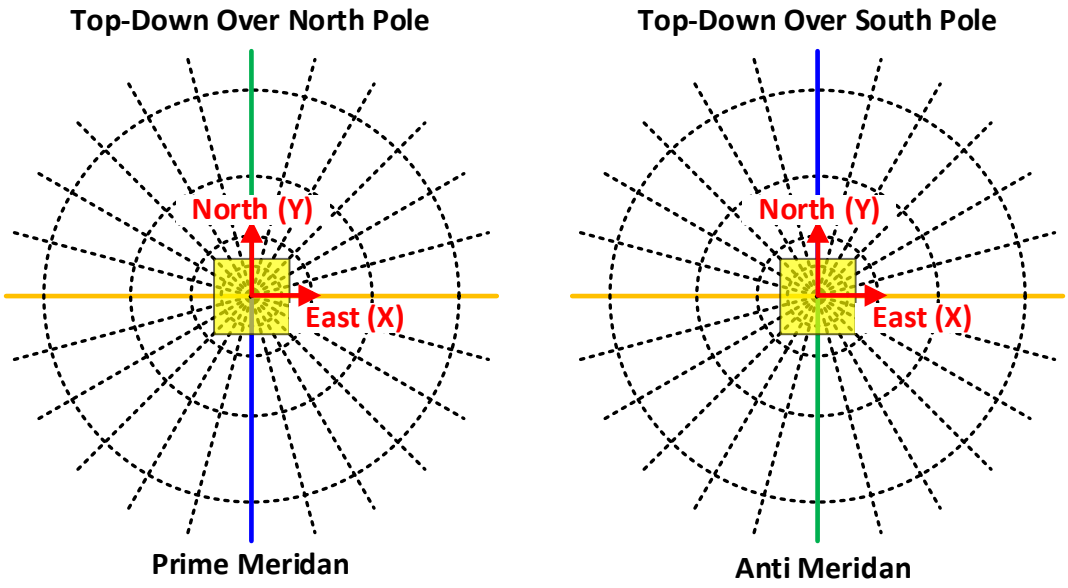


Figure 8: ENU-LRP over the North and South Poles

When absolute geodetic stages move toward either pole, the ENU-LRP orientation will always have the north axis pointing toward the North Pole; however, when passing over either pole the ENU-LRP orientation will suddenly rotate to match these specific over-the-pole definitions.

This standard represents a stage's absolute geodetic orientation using "ENU", above a Q, i.e., ${}^{ENU}Q_i\ (\alpha, \beta, \psi)$, where Q_i is the orientation for the i^{th}stage with α, β, and ψ specifying the rotations around the ENU coordinate system at a given absolute position. Specifically, α is the rotation around the Z-axis (Up-axis), β is the rotation around the new Y-axis (new North-axis), and ψ is the rotation around the new-new X-axis (new-new East-axis), in that order. The rotations are the angles necessary to rotate the parent stage's orientation to the absolute orientation, or if the absolute stage is the root stage the angles are how the absolute orientation measurement device is affixed to the hosting object (e.g., platform).

For example, Equation 4 is the absolute geodetic orientation for an aircraft heading due east, with no pitch or roll measurement.

$$^{\text{ENU}}_{}Q_i\ (\alpha, \beta, \psi) = {}^{\text{ENU}}Q_i\ (0°, 0°, 0°) \qquad \textbf{Equation 4}$$

For example, Equation 5 is the absolute geodetic orientation for an aircraft heading west, i.e., 270° compass heading, pitched up 15°, with a 10° clockwise roll (as if sitting in the cockpit looking forward). Equation 5 states the orientation has a $180°$ rotation around the Z-axis (from the east), a $-15°$ rotation around the new Y-axis (now pointing south), and a $10°$ roll around the new-new X-axis.

$$^{\text{ENU}}_{}Q_i\ (\alpha, \beta, \psi) = {}^{\text{ENU}}Q_i\ (180°, -15°, 10°) \qquad \textbf{Equation 5}$$

When an absolute position is known for the stage, that position defines the orientation of the ENU axis; it is possible to measure the ENU-LRP orientation without knowing the position.

6.1.2.1.2 Absolute Geocentric Orientation

The Staging System's absolute geocentric orientation is a set of rotations around the ECEF axis, for any point in the ECEF system. Unlike the absolute geodetic orientation, the ECEF orientation is the same for every point in the ECEF space.

This standard represents a stage's absolute geocentric orientation using "ECEF", above a Q, i.e., $\overset{ECEF}{Q_i}\ (\alpha, \beta, \psi)$, where Q_i is the orientation for the i^{th}stage with α, β, and ψ specifying the rotations around the ECEF coordinate system. Specifically, α is the rotation around the earth's Z-axis, β is the rotation around the new Y-axis, and ψ is the rotation around the new-new X-axis, in that order.

Figure 9 illustrates the Absolute Geocentric Orientation at the point in the middle of the yellow plane. The dashed black line shows the point's location relative to the earth. The yellow plane's green axes align with the earth's ECEF axes. The yellow plane's red axes are the offsets from the ECEF axes, specifically α is the rotation around the Z axis and ψ is the rotation around the X axis.

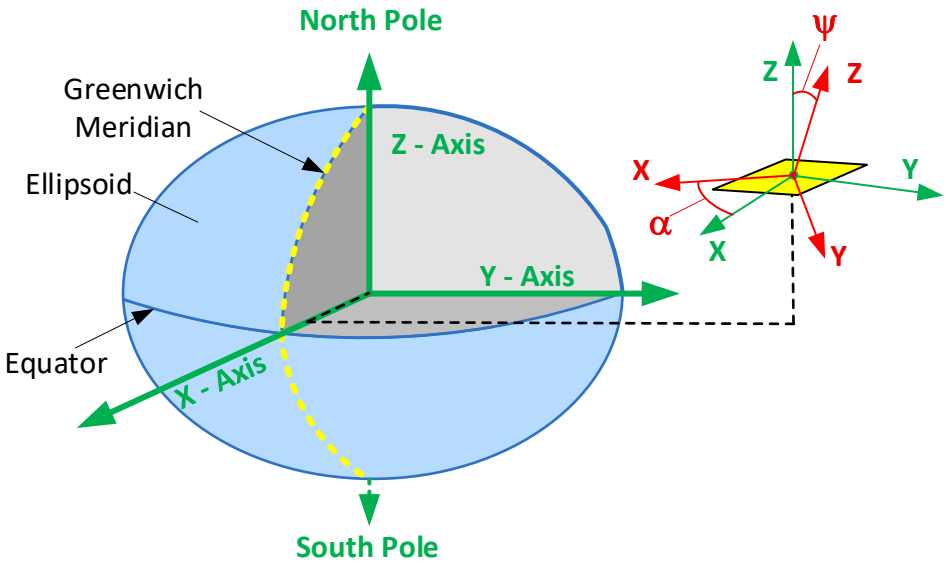


Figure 9: Absolute Geocentric Orientation

6.1.2.2 Relative Orientation

The Staging System's relative orientations measure rotations as offsets to the stage's parent stage orientation. This standard represents a stage's relative orientation using an arrow arc ($\curvearrowright$) above the letter Q, i.e., $\widetilde{Q}_i(\alpha, \beta, \psi)$, where Q_i is the orientation for the i^{th}stage with α, β, and ψ specifying the rotations around the parent stage's (S_{i-1}) coordinate system. Specifically, α is the rotation around the parent's Z-axis, β is the rotation around the new Y-axis, and ψ is the rotation around the new-new X-axis, in that order.

6.1.2.3 Orientation Matrices

Stage orientations provide the information necessary to construct rotation matrices. Equation 6 illustrates the matrix notation for a single axis rotation, where the subscript defines the axis of rotation (Z in this example), and the parenthesized value defines the amount of rotation (α in this example).

$$R_Z(\alpha) \qquad \textbf{Equation 6}$$

Equation 7 defines a stage's three axes rotation in matrix form. The Staging System uses intrinsic and alibi operations (see Appendix C) so the matrix order is, R_Z is first, R_Y is second, and R_X is last. When operating on a vector, or a point in 3-space, the alibi operation means the vector will be post-multiplying the rotation operation.

$$\widetilde{Q_i}(\alpha,\beta,\psi) = R_Z(\alpha) * R_Y(\beta) * R_X(\psi) \qquad \textbf{Equation 7}$$

Equation 8 expands the definition of each rotation matrix.

$$\widetilde{Q_i}(\alpha,\beta,\psi) = \begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\beta & 0 & \sin\beta\\ 0 & 1 & 0\\ -\sin\beta & 0 & \cos\beta\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\psi & -\sin\psi\\ 0 & \sin\psi & \cos\psi\end{bmatrix} \qquad \textbf{Equation 8}$$

Equation 9 shows the final product of all three matrices from Equation 8.

$$\begin{bmatrix}\cos\alpha\cos\beta & \cos\alpha\sin\beta\sin\psi - \cos\psi\, sin\,\alpha & \sin\alpha\sin\psi + \cos\alpha\, cos\,\psi\sin\beta\\ \cos\beta\sin\alpha & \sin\psi\sin\beta\sin\alpha + \cos\psi\cos\alpha & \cos\psi\sin\beta\sin\alpha - \sin\psi\cos\alpha\\ -\sin\beta & \sin\psi\cos\beta & \cos\psi\cos\beta\end{bmatrix} \qquad \textbf{Equation 9}$$

Equation 10 is an alternate representation for Equation 9 to condense the size on the page ($r_{1,1} = cos\beta cos\alpha, etc.$).

$$\begin{bmatrix}r_{1,1} & r_{1,2} & r_{1,3}\\ r_{2,1} & r_{2,2} & r_{2,3}\\ r_{3,1} & r_{3,2} & r_{3,3}\end{bmatrix} \qquad \textbf{Equation 10}$$

An absolute geodetic orientation, $\overset{ENU}{Q_i}(\alpha,\beta,\psi)$, is the combination of two rotation matrices: ENU rotation and an "inceptive" rotation. The ENU rotation is the matrix necessary to rotate from the ECEF XYZ coordinate system to the ENU coordinate system for a given absolute position, i.e., $\overset{XYZ\rightarrow ENU}{R}(\phi_s,\lambda_s)$, see Appendix C. The inceptive, or beginning, rotation is an optional relative rotation from the parent stage's orientation to the absolute geodetic orientation. The inceptive rotation is the same as a relative stage rotation matrix, i.e., Equation 7. Equation 11 computes the absolute geodetic orientation rotation matrix.

$$\overset{\text{ENU}}{Q_i}(\alpha,\beta,\psi) = \overset{XYZ\rightarrow ENU}{R}(\phi_s,\lambda_s) * R_Z(\alpha) * R_Y(\beta) * R_X(\psi) \qquad \textbf{Equation 11}$$

By default, an absolute geodetic orientation with all three rotation values set to zero, i.e., $\overset{\text{ENU}}{Q}_1(0,0,0)$ means the X-axis points due east, the Y-axis points north, and the Z-axis points up.

A constellation typically defines an absolute geodetic orientation for the root stage followed by one or more relative orientations. Figure 10 illustrates a constellation with three stage orientations for a ground scenario:

1) An absolute stage, $\overset{ENU}{Q}_1(0,0,0)$, from a GPS antenna phase center; the zero values mean the axes align with East, North, and up
2) A relative stage for the platform at the vehicle center of gravity, $\widehat{Q}_2(\alpha_2, \beta_2, \psi_2)$
3) A relative sensor stage, $\widehat{Q}_3(\alpha_3, \beta_3, \psi_3)$, showing the sensor pointing vector using the Geo-Intelligence Sensor Common Pointing Orientation (GSCPO), see Section 6.5.

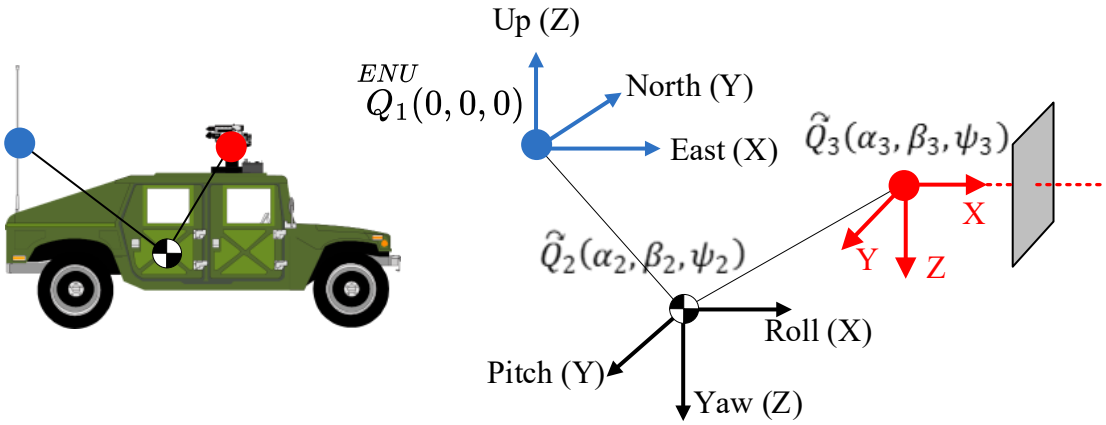


Figure 10: Orientation Example

If a constellation were to have a relative orientation root stage, the constellation would define a local orientation system without an absolute reference; however, a latter stage can introduce an absolute reference. All child stages of the constellation resolve to the orientation of the root stage.

6.1.3 Future updates

The Staging System modeling plans include adding additional coordinate and orientation methods. The use of Earth-Centered Inertial (ECI) coordinates for satellites or other earth orbiting objects may be included in future revisions of this standard; this includes metadata documented in MISB ST 1504 - Natural Representation of Orbital State Vectors [4].

Future versions of this standard may include the use of quaternions as an alternate method of specifying orientations of stages. Some of the modeling attributes have reserved identifiers for the addition of quaternion kinematic information.

6.2 Staging System Mathematics

The mathematical goal of the Staging System is to resolve positions and orientations of any stage in a constellation to its root stage's coordinate system. The staging system provides both absolute and relative information. Readers resolve an absolute position by combining the relative information into a single mathematical transformation together with the absolute information.

6.2.1 Relative Stage Mathematics

Each relative stage defines the position and orientation metadata, i.e., $S_i = (P_i, Q_i)$, enabling the construction of a 4x4 rotation/translation matrix. Equation 12 shows each relative stage, S_i, has two categories of components, the position offsets from the parent stage ($\vec{P}$ or $\measuredangle{P}$) and the rotations relative to the parent stage ($\widetilde{Q}$), the attributes of each combining to define the relative stage 4x4 matrix. Convert all $\measuredangle{P}$ to their equivalent $\vec{P}$ values (see Equation 1) before creating the following matrix.

$$S_i = \left(\vec{P}_i, \widetilde{Q}_i\right) = \begin{bmatrix} r_{1,1} & r_{1,2} & r_{1,3} & x \\ r_{2,1} & r_{2,2} & r_{2,3} & y \\ r_{3,1} & r_{2,3} & r_{3,3} & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{Equation 12}$$

Combining adjacent relative stage matrices, by multiplication, produces a single transformation matrix, which this document labels as a "full" matrix. A full matrix transforms locations in the last stage's coordinate system into positions relative to the first stage. Equation 13 shows an example matrix product of a list of stage matrixes (in order) producing a *full* stage matrix enabling transformations of coordinates in the terminal stage into coordinates relative to the first relative stage. This example shows the stage matrices for one "branch" of the whole constellation, i.e., Stage 2 connects to Stage 4 which connects to Stage 7, etc. to the terminal Stage.

$$S_{full} = S_2 S_4 S_7 \dots S_{terminal} = \prod_{i=\{2,4,7,\dots,terminal\}} S_i \quad \text{Equation 13}$$

Equation 14 continues this example and shows the transformation of a point relative to the terminal stage ($\boldsymbol{p}_{terminal}$) to a point relative to Stage 2 ($\boldsymbol{p}_2$). Since the S matrixes are 4x4, each point needs to include an extra dimension with a value set to 1. Equation 15 shows the points including this extra dimension. Typically, a fourth dimension with a point defines a homogeneous point for a projective system which is not the intent in the Staging System; the extra dimension makes the math correct for these computations and the reported value does not include it.

$$\boldsymbol{p}_2 = S_{full} \boldsymbol{p}_{terminal} \quad \text{Equation 14}$$

$$\begin{pmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{pmatrix} = S_{full} \begin{pmatrix} x_{terminal} \\ y_{terminal} \\ z_{terminal} \\ 1 \end{pmatrix} \quad \text{Equation 15}$$

Figure 11 shows an example of four stages with the root stage at the origin and stage 4 as the terminal stage, which is oriented in the GSCPO (see Section 6.5). The figure shows an example data point, P (in blue), with coordinates of (0, 0, 2) relative to the terminal stage, i.e., 2 meters along the x-axis. Equation 13 enables the computation of the blue point's coordinates relative to the root stage (P_{root}) using the multi-stage transformation (S_{full}) and the coordinates of P.

List of stages in the format of:
$S(x, y, z, \alpha, \beta, \psi)$

S_1(0, 0, 0, 0, 0, 0)

S_2(2, 3, 0, 0, 0, 0)

S_3(0, 0, 3, **90**°$_Z$, 0, **180**°$_X$)

S_4(2, 0, 0, **125**°$_Z$, **-35**°$_Y$, 0)

$$P = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \text{(Relative to Stage 4)}$$

$$P_{root} = S_{full} * P = S_2 * S_3 * S_4 * P$$

$$\mathrm{P_{root}} = \begin{pmatrix} 3.4091 \\ 4.0133 \\ 1.7955 \end{pmatrix}$$

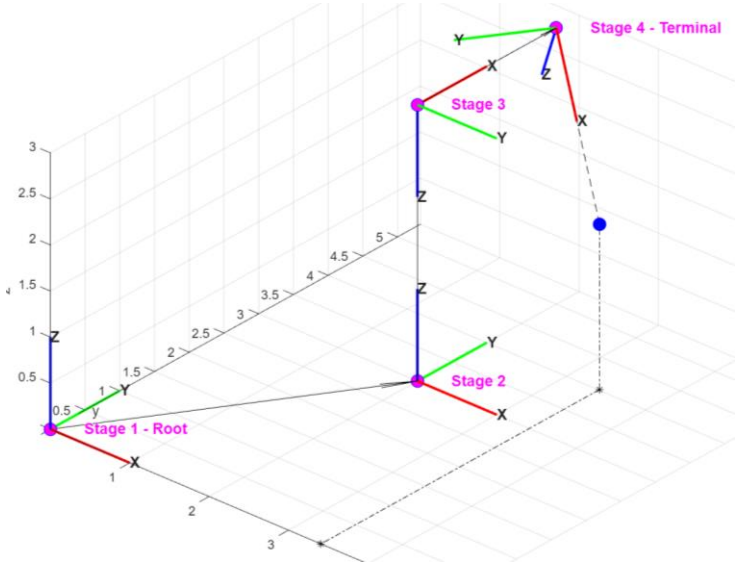


Figure 11: Example of Four Stage Constellation with Sample Point

6.2.2 Absolute Stage Mathematics

There are two types of absolute stages, geocentric (ECEF) and geodetic. A geocentric stage extends the relative stage's mathematics to produce absolute positions. The geodetic stage has additional steps to produce absolute positions.

6.2.2.1 Geocentric Stage Mathematics

A geocentric absolute stage uses the same reference frame information as a relative stage, i.e., meters in a 3-D Cartesian coordinate system. Therefore, computing the absolute coordinate of a point relative to a geocentric absolute stage is the inclusion of the absolute stage product to the front of Equation 14, as shown in Equation 16. Creating the S_{ECEF} matrix uses the same processes as a relative stage in Equation 12. The result of Equation 16 is $\mathbf{P_{ECEF}}$ which is the absolute position of the $\mathbf{P_{terminal}}$ point in ECEF coordinates.

$$\boldsymbol{p}_{ECEF} = S_{ECEF} S_{full} \boldsymbol{p}_{terminal} \qquad \textbf{Equation 16}$$

6.2.2.2 Geodetic Stage Mathematics

A geodetic absolute stage uses Latitude, Longitude, and HAE as part of its reference frame information, which is different than a relative stage; therefore, computing absolute geodetic coordinates of a relative point includes additional steps. The first step is to apply the rotational component of the geodetic stage to Equation 14. The second step converts the $\boldsymbol{p}_{root}$ to geocentric, adds the $\boldsymbol{p}_{\boldsymbol{geoRel}}$ position, then converts back to geodetic coordinates.

For the first step, Equation 17 defines the S_{rotation} matrix, which is the rotational part of the geodetic absolute stage, formed from α, β, ψ using Equation 10. Equation 18 shows the

application of the rotation matrix to the front of Equation 14, producing $\mathbf{P}_{\mathbf{geoRel}}$, which is a point relative to the geodetic absolute stage.

$$S_{rotation} = \begin{bmatrix} r_{1,1} & r_{1,2} & r_{1,3} & 0 \\ r_{2,1} & r_{2,2} & r_{2,3} & 0 \\ r_{3,1} & r_{3,2} & r_{3,3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \textbf{Equation 17}$$

$$\boldsymbol{p}_{geo\,\mathrm{Re}\,l} = S_{Rotation} S_{full} \boldsymbol{p}_{terminal} \quad \textbf{Equation 18}$$

Equation 19 shows the second step, first transforming $\boldsymbol{p}_{root}$ to ECEF, then adding the transformed point to $\boldsymbol{p}_{geoRel}$ and finally converting the resulting coordinate back to geodetic. The functions f and g are the conversion to ECEF and back to geodetic as shown in Appendix B

$$\boldsymbol{p}_{geodetic} = g\left(f(\boldsymbol{p}_{root}) + \boldsymbol{p}_{geoRel}\right) \quad \textbf{Equation 19}$$

Alternatively, systems may use algorithms to compute the position directly, without the transformation into ECEF.

6.2.3 Reverse methods

The staging system defines each stage as changes to the previous stage allowing consistency with the mathematics for working with the stages. With certain stage configurations using absolute position or orientation, there are cases where the absolute stage computation needs to compute its position or orientation backwards, or in reverse, through the staging system. The result is the preceding stage's position and/or orientation becomes absolute. The process of reversing the computation is unique for positions and orientations.

6.2.3.1 Reverse Position

When a child stage with an absolute position is based on its parent's stage with a relative position, there may be a need to compute a reverse position. By definition, a child stage with an absolute position is co-located with the parent's stage; the two positions are at the same location but use different coordinate systems to represent those positions. The goal is to have the parent (and potentially the parents of the parent) assume the absolute coordinate system. For example, the left side of Figure 12 illustrates 2-D staging positions starting with the Root position defining a relative origin; P_1 is at a relative offset from the Root; P_2 is at a relative offset from P_1; P_3 is at a relative offset from P_2; and P_4 is an absolute position (relative to the Earth) which is co-located with P_3. The right side shows the information (data structure) organization of the stages. The goal in the example is to define the absolute coordinates of the root position, so all relative stage children can compute absolute coordinates if necessary.

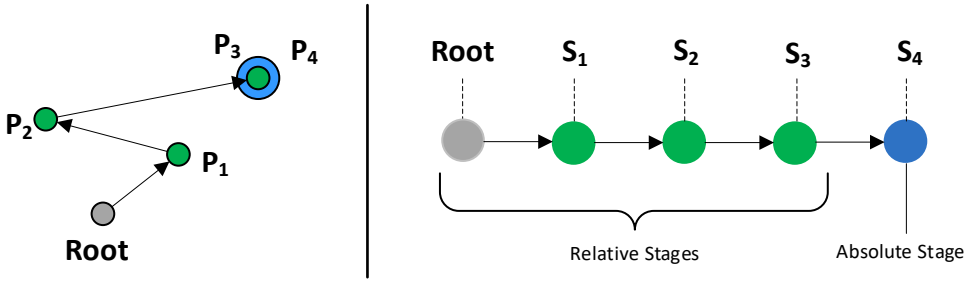


Figure 12: Staging Example for Reverse Position Computation

The reverse position computation has two steps: consolidate relative stages and perform coordinate conversion. Readers consolidate relative stages by creating a *full* matrix for all the desired parent stages using the steps in Section 6.2.1. In the preceding example a reader would combine stages, root, 1, 2, and 3, into the full matrix. If the absolute position uses ECEF coordinates, the result is a vector between the root and the absolute stage, and the absolute position is a vector from the center of the earth to the absolute position. To perform the coordinate conversion, convert the absolute position to ECEF (if necessary) and subtract the vector between the root and absolute position from the ECEF vector of the absolute position, resulting in the absolute position of the root. Figure 13 illustrates the coordinate conversion for the Figure 12 example, with the absolute coordinates of P_4 defining a vector $\vec{P}_4$ and the consolidated relative stages defining the $\overrightarrow{Root}$ vector. The coordinates of the Root position is the value of $\overrightarrow{Root}_{Absolute} = \vec{P}_4 - \overrightarrow{Root}$.

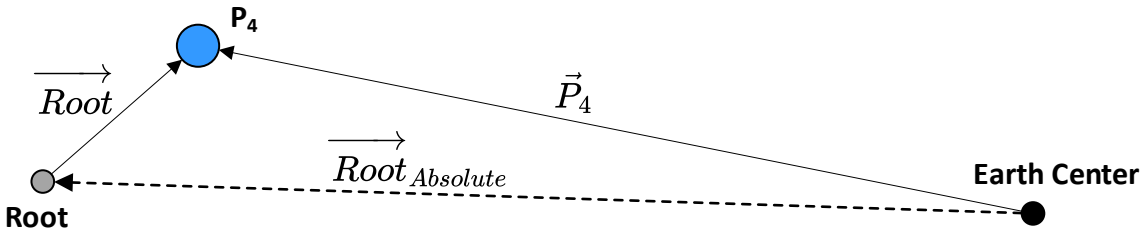


Figure 13: Reverse Position Coordinate Conversion

After determining the absolute position of the Root, readers may update the root stage's 4D matrix (Equation 12) to include the new position. With the updated matrix, all child stages which relate to the root can compute absolute coordinates when performing the matrix multiplications. In the Figure 12 example Stages 1 through 3 can now determine their absolute positions along with any other relative stages which may depend on the root, which are not shown in the figure.

6.2.3.2 Reverse Orientation

When a child stage with an absolute orientation is based on a parent's stage with a relative orientation, a reader may need to compute a reverse orientation. A stage with an absolute orientation provides the rotations from the parent stage's axes to the absolute orientation's axes; these rotations perform as a relative stage. The goal is to have the parent (and potentially the parents of the parent) assume the absolute orientation from the absolute stage, thereby providing

the absolute orientation for all stages based on the highest parent, e.g., the root. For example, the left side of Figure 14 illustrates an absolute orientation Q_3 (blue) based on the relative Q_2 orientation; the Q_2 orientation based on the relative Root orientation; and the Q_1 orientation based on the Root orientation. The right side shows the information (data structure) organization of the stages, specifically, S_1, is not in the linage of the S_2 branch. The goal in the example is to define the absolute orientation of the Root, so all relative stage children can compute their absolute orientations if necessary.

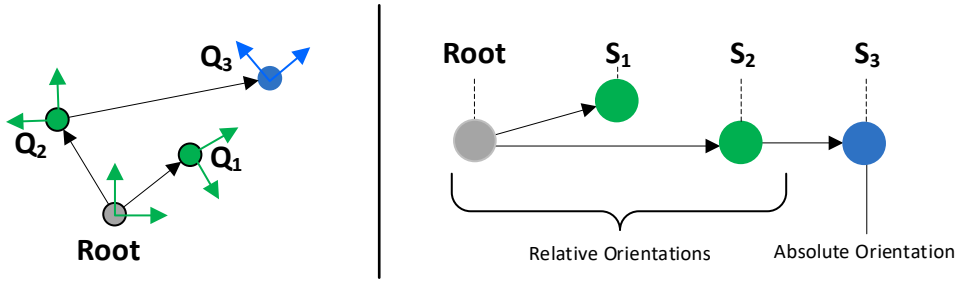


Figure 14: Reverse Orientation Conversion

The reverse orientation computation has two steps: consolidate relative stages and perform orientation conversion. Readers consolidate relative stages by creating a *full* matrix for all the desired parent stages using the steps in Section 6.2.1. In the preceding example a reader would combine stages, Root, 2, and 3, into the full matrix. The result is the S_{full} matrix which includes the rotations to rotate the Root's axes to the absolute axes. To perform the orientation conversion, readers need to compute the rotations from the absolute axes to the Root's axes, which is the mathematical inverse of the S_{full} matrix, i.e., $S_{reverse} = S_{full}^{-1}$. The left side of Figure 15 illustrates a visual example from Figure 14, showing the arrows (red) from the absolute stage to the root reverse direction, providing a new pseudo-root starting at the absolute stage. The right side of the figure shows the staging result of consolidating the relative stages and computing the reverse orientation.

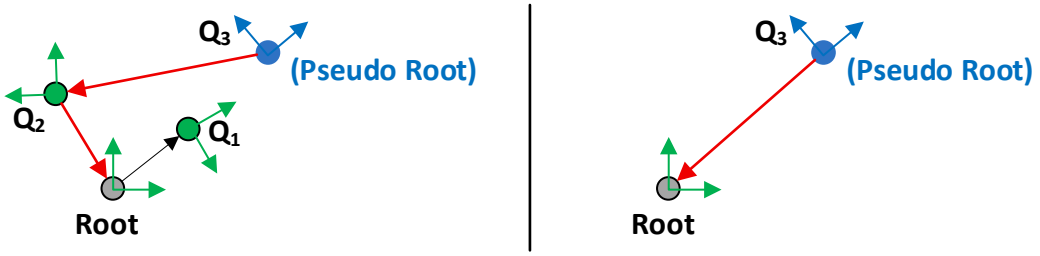


Figure 15: Reversed Orientation

After determining the "pseudo-root", readers use substitute S_{ECEF} with $S_{reverse}$ in Equation 16. With the updated matrix, all child stages which relate to the root can compute absolute orientations when performing the matrix multiplications.

The reverse orientation provides the absolute orientation of the Root but this orientation is relative to the position of the pseudo root, i.e., the ENU-LRP of the Root will not be co-planar with the Pseudo-Root unless the position of the Root and Pseudo-Root are the same. These are optional adjustments depending on the readers' use case and how close the Root and Pseudo-Root are located to each other.

6.3 Stage Configurations

The purpose of the staging system is to enable systems to describe the physical locations and orientations of their devices with as much detail as necessary to achieve the systems objectives. Devices can include GPS, INS, IMU, actuators, gimbals, accelerometers, sensors, lever arms, and more. These devices may be in any configuration and can measure both absolute and relative values. A configuration is how a producer links the different stages together and if/where the producer puts relative or absolute stages. The following sub-sections describe three example configurations: Linear, Forked, and Single Stage.

6.3.1 Linear Configurations

Figure 16 illustrates a linear stage configuration example, starting with a Root Stage, then Stage 2 is based on the Root Stage, then Stage 3 is based on Stage 2, etc., ending with the Terminal Stage, based on Stage 4.

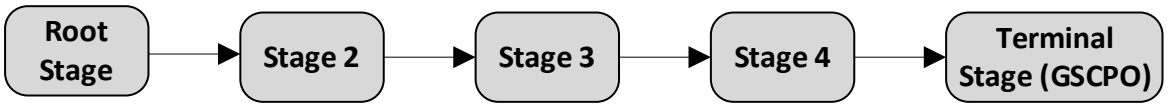


Figure 16: Stage Configuration Example

If all the stages are relative stages (i.e., both positions and orientation are relative), the terminal stage GSCPO (see Section 6.5) pointing vector will be relative to the orientation of the root stage. A use case example for an all-relative configuration, is a robot inside a building without GPS; it may not know its world-based position but can accurately determine its relative position inside the building and its pointing angles.

If any of the stages use an absolute position, then the whole configuration's positions are determinable per the equations in Section 6.2. When a stage's absolute position is known, both the location on the earth and the orientation relative to the earth are known. For absolute geodetic positions, the ENU-LRP at that position becomes the orientation for latter stages. For absolute geocentric positions, the ECEF cartesian axis becomes the orientation for the latter stages, but with the origin at the given position. For example, if Stage 2 is the only stage with an absolute position, Stages 3 through the Terminal stage's absolute position and orientation are computable using the techniques in Section 6.2.2. The absolute position of the Root Stage is computable using techniques in Section 6.2.3.

When two or more stages in the configuration define the absolute position, it is up to the reader to resolve which position to use, depending on device accuracy, etc. The expectation is the absolute stage closest to the terminal stage is the stage from which to base the pointing orientations.

6.3.2 Forked Configuration

Figure 17 illustrates a forked stage configuration example, with a Root Stage, then stages 2, 4, and 5 are all based on the Root Stage. In this illustration, the blue stages are absolute; Stage 3 defines an absolute position and Stage 4 defines an absolute orientation. A use case for this configuration is an aircraft, the Root Stage is positioned at a center reference point, e.g., center of gravity (CG), the second stage is the offset to the GPS antenna, the third stage reports the GPS coordinates, the fourth stage is offset to the IMU and reports the roll, pitch, and yaw of the aircraft, and stage 5 is the offset to the sensor stage.

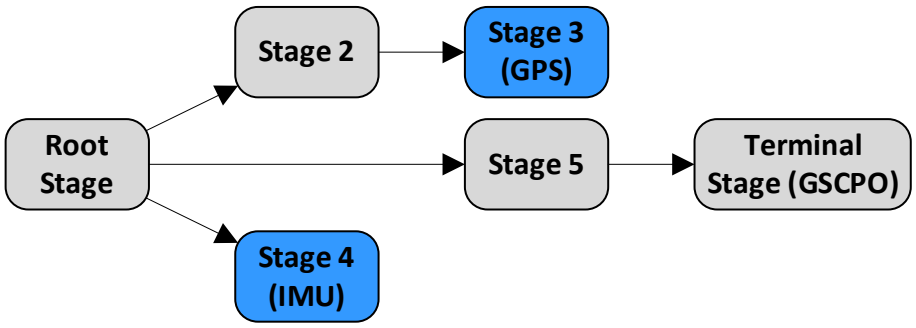


Figure 17: Stage Configuration Example 2

With this configuration, the first step in computing the GSCPO pointing direction is to determine the absolute position and orientation of the root stage. Since Stage 3 is an absolute position with Stage 2 as the parent, the physical location in space is the same place but stated with different coordinate systems, Stage 2 is relative coordinates from the Root Stage and Stage 3 is absolute coordinates on the earth. Using the techniques in Section 6.2.3, the absolute position of the Root Stage is computable from the offsets for Stage 2 and the position information in Stage 3.

The Stage 4 IMU may be physically located away from the Root Stage by some relative offset, but the orientation of both stages, relative to Stage 4's ENU-LRP, is the same. However, with the separation, the ENU-LRP at each stage is slightly different, depending on the distance. The reader can either use Stage 4's orientations at the Root Stage or adjust them according to the change in the absolute position.

With both the absolute position and orientation, the reader uses the techniques in Section 6.2.2 to determine the Terminal Stage's pointing direction.

6.3.3 Single Stage Configuration

Figure 18 illustrates a Single Stage Configuration, where a single stage defines both position and orientation to the GSCPO pointing vector of the sensor.

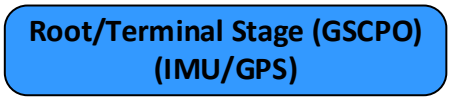


Figure 18: Single Stage Configuration Example

In this example, the system may have one IMU/GPS device to define the staging values, or the producer may combine the information from several devices into one stage. The goal of the

MIMD model and staging system is to describe and record the true configuration and metadata from systems so downstream analysis has as much information about the system as possible for performing analysis.

6.4 MIMD Modeling the Staging System

To include the Staging System into the MIMD Model as a hierarchy causes a conflict with the MIMD Model's hierarchical architecture. Using a different representation for the Staging System provides a format which merges with the MIMD Model hierarchy while retaining the Staging System hierarchical information.

This alternative representation converts the stage hierarchy to a linked list. Each stage includes a unique numerical identifier, the mimdId (see MISB ST 1901 [5]), and a parent identifier (parentStage). The parentStage refers to the mimdId of the stage's parent. The parentStage values are always greater than zero unless the stage is the root stage, in which case the parentStage is zero indicating the root stage does not have a parent.

Table 2 shows a comparison of the hierarchical Staging System from Table 1, on the left, and the linked list Staging System, on the right. On the right, the stages are a list, with each stage numbered, containing a parentStage, and includes the position/orientation data. In this example the stage number is the stage identifier and the parentStage is the stage identifier of the parent stage. Since the first element in the list is the root, the parentStage is set to zero. The second element in the list shows the parentStage=1, which is the root stage. This pattern continues for the whole list. The Staging System description in both sides of the table is identical.

Table 2: Example Staging System Metadata Representation

Stage 1 - Reference Frame Survey Point Absolute Position/Orientation Stage 2 – Trailer Relative Position / Orientation Stage 3 – Left Camera Relative Position / Orientation Stage 4 – Right Camera Relative Position / Orientation Stage 5 – Top Center Gimbal Relative Position / Orientation Stage 6 – Left Upper Camera Relative Position / Orientation Stage 7 – Right Upper Camera Relative Position / Orientation	Stage 1 – Reference Frame Survey Point parentStage = 0 Absolute Position/Orientation Stage 2 – Trailer parentStage = 1 Relative Position / Orientation Stage 3 – Left Camera parentStage = 2 Relative Position / Orientation Stage 4 – Right Camera parentStage = 2 Relative Position / Orientation Stage 5 – Top Center Gimbal parentStage = 2 Relative Position / Orientation Stage 6 – Left Upper Camera parentStage = 5 Relative Position / Orientation Stage 7 – Right Upper Camera parentStage = 5 Relative Position / Orientation

The advantage of the list method is the individual list elements can distribute throughout the whole MIMD Model. This means MIMD Model objects (e.g., platform or sensor) do not require any additional staging information beyond their own relationship with their parents. Additionally, the list method enables reporting staging information at different rates within the modeling data stream. This allows dynamic gimbals to update frequently and fixed mounts to update infrequently, resulting in more efficient use of bandwidth.

At various places within the MIMD Model, some classes will have a list of Stages, e.g., the Payload class has a stages attribute. Because lists can have multiple stages, the class hosting the list may have an attribute to associate a particular stage with that device. For example, the ST 1907 [6] Geo-Intelligence Sensor class provides a *gscpoStage* attribute to indicate which stage is the GSCPO "pointing stage" for the sensor.

6.5 Geo-Intelligence Sensor Common Pointing Orientation (GSCPO)

When specifying the Staging orientation of any Single Point or Multi-Point Geo-Intelligence Sensor (see MISB ST 1907) the MIMD Model defines a **Geo-Intelligence Sensor Common Pointing Orientation (GSCPO)** for the terminal stage of where the sensor is pointing. In all cases (Single Point or Multi-Point) the GSCPO has the positive x-axis pointing in the direction from the sensor to the Scene or point-target. The x-axis aligns with the boresight of the pointing sensor for the given sensor. With Multi-Point Sensors an array of samples is available (either one or two dimensional), so there is enough information to define a Scene and therefore the y and z axes orientations. The positive y-axis points to the right side of the Scene or in the direction of

increasing column numbers. The positive z-axis points downward towards the bottom of the Scene or in the direction of increasing row numbers. Figure 19 illustrates the final two stages of a stage list, where S_n is the terminal stage. The orientation of S_n's x-axis is the direction to the point-target or Scene, depending on the type of geo-intelligence sensor.

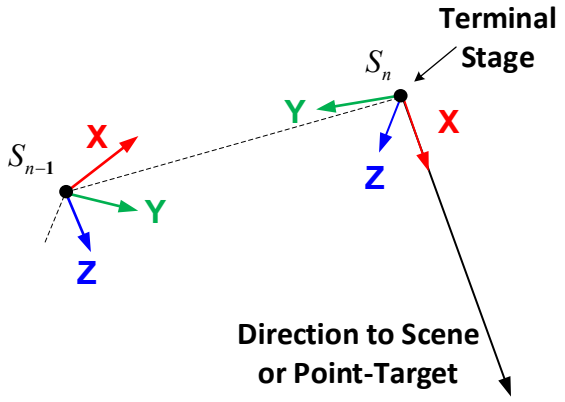


Figure 19: Geo-Intelligence Sensor Common Pointing Orientation (GSCPO)

6.6 Additional Stage Information

A Stage class includes the MISB ST 1904 Base Attributes; therefore, positions and orientation may include timer specifications with timer offsets and uncertainty information. Devices which report staging metadata may include their own timer or have a reporting delay from an existing timer; in these cases, the timerId and timerOffset values are set.

These devices may include uncertainties (e.g., encoder errors, gimbal axis non-orthogonality, bearing wobble, etc.). The numericalPrecision attribute allows for this additional metadata in the form of standard deviations and correlation coefficients.

7 Model Classes

7.1 Stage Class

Figure 20 shows the UML class diagram for the MIMD Staging System. The Stage class provides the identity of the stage and its reference to the parent. Each Stage class has a Position and Orientation. The Position is either an Absolute Geocentric, Absolute Geodetic, or Relative. A Position may include kinematics information which supports velocity, acceleration, and other levels of kinematics. The Orientation is either an Absolute ENU orientation or Relative orientation. An orientation may include kinematics information which supports angular velocity, acceleration, etc.

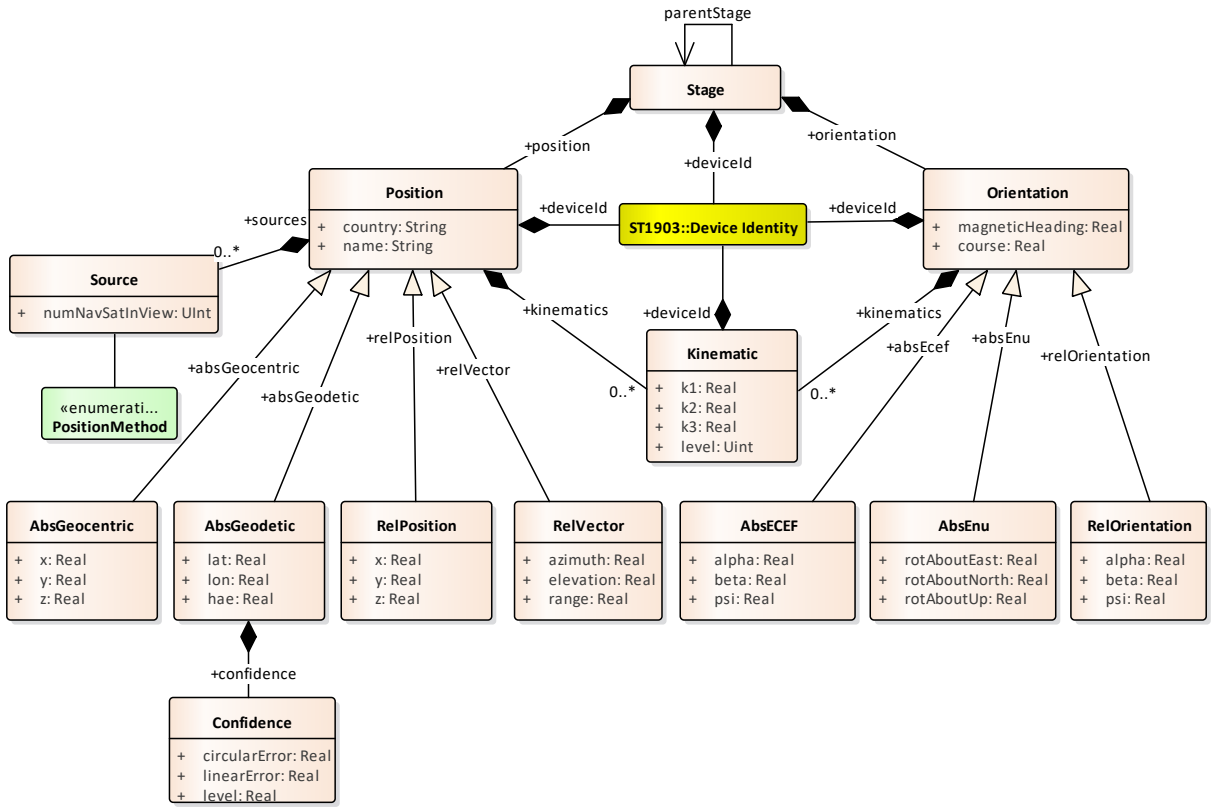


Figure 20: Staging System Class Model

Per discussion in MISB ST 1901, MIMD classes support the Report-On-Change (reference the Motion Imagery Handbook [7]) and include MIMD Base Attributes [8], which defines attribute identifiers 1-32.

The Stage class contains attributes for the parent identifier, a Position class, and an Orientation class for a stage's position.

Table 3: Stage Class Attributes[1]

[1] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

Id	Name	Type	Min	Max	MLen	Units	Ref
33	parentStage	REF<Stage>	N/A	N/A	N/A	None	7.1.1
34	position	Position	N/A	N/A	N/A	None	7.1.2
35	orientation	Orientation	N/A	N/A	N/A	None	7.1.3
36	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.1.4

7.1.1 Attribute 33 – parentStage

This attribute is a directed association (or reference) to Class Stage. Section 7.1 defines the Stage class.

The *parentStage* attribute is a reference to the stage's parent. Each stage instance must include both a mimdId and a parentStage attribute value. The mimdId and parentStage may employ Report-on-Change techniques to conserve bandwidth. A parentStage reference set to zero indicates the stage is a root stage and does not have a parent.

Requirement(s)	
ST 1906-01	A Stage class instance mimdId attribute value shall never be "Unknown" (ROC_Unknown).
ST 1906-02	A Stage class instance parentStage attribute value shall never be "Unknown" (ROC_Unknown).

7.1.2 Attribute 34 – position

This attribute is a composite relationship to the Class Position. Section 7.2 defines the Position class.

7.1.3 Attribute 35 – orientation

This attribute is a composite relationship to the Class Orientation. Section 7.7 defines the Orientation class.

This attribute is a composite relationship to the Class Orientation

7.1.4 Attribute 36 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.2 Position Class

The Position class contains attributes (see Table 4) for the country, the method used to determine the position, the position values which inherit from one of the three child classes (AbsGeocentric, AbsGeodetic, or RelPosition) and a *Kinematic* class list.

Table 4: Position Class Attributes[2]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	country	String	N/A	N/A	100	None	7.2.1
34	absGeocentric	AbsGeocentric	N/A	N/A	N/A	None	7.2.2
35	absGeodetic	AbsGeodetic	N/A	N/A	N/A	None	7.2.3
36	relPosition	RelPosition	N/A	N/A	N/A	None	7.2.4
37	kinematics	LIST<Kinematic>	--	--	N/A	None	7.2.5
38	sources	LIST<Source>	--	--	N/A	None	7.2.6
39	relVector	RelVector	N/A	N/A	N/A	None	7.2.7
40	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.2.8
41	name	String	N/A	N/A	100	None	7.2.9

7.2.1 Attribute 33 – country

The *country* attribute value is a string containing the three-character short URN-based Identifier country code format defined in GENC [9]. The short URN-based Identifier country code supports both GENC and ISO3166 country codes. For example, the GENC country code for the United States is "ge:GENC:3:3-17:USA" and for ISO 3166 its "ge:ISO1:3:VII-19:USA".

Systems producing MIMD may use GENC or ISO 3166 country codes.

Requirement(s)	
ST 1906.3-11	Producers shall format the Position class instance's country attribute using the three-character short URN-based country code format per GENC.
ST 1906.3-12	United States systems shall use GENC country codes.

7.2.2 Attribute 34 – absGeocentric

This attribute is a specialization relationship to the Position class. Section 7.3 defines the AbsGeocentric class.

7.2.3 Attribute 35 – absGeodetic

This attribute is a specialization relationship to the Position class. Section 7.4 defines the AbsGeodetic class.

[2] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

7.2.4 Attribute 36 – relPosition

This attribute is a specialization relationship to the Position class. Section 7.6 defines the RelPosition class.

7.2.5 Attribute 37 – kinematics

This attribute is a list of Kinematic classes. Section 7.10 defines the Kinematic class.

7.2.6 Attribute 38 – sources

This attribute is a list of Source classes. Section 7.11 defines the Source class.

7.2.7 Attribute 39 – relVector

This attribute is a specialization relationship to the Position class. Section 7.12 defines the RelVector class.

7.2.8 Attribute 40 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.2.9 Attribute 41 – name

The *name* attribute is the operational name of the Position. The operational name supplies a short description or name for the Position, e.g., "Washington Monument". The operational name is not the manufacturer's name of the Position device. See the deviceId attribute to provide information about the Position device.

7.3 AbsGeocentric Class

The AbsGeocentric class has attributes (see Table 5) for an Absolute ECEF position.

Table 5: AbsGeocentric Class Attributes (extends Position)[3]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	x	Real	--	--	N/A	m	7.3.1
34	y	Real	--	--	N/A	m	7.3.2
35	z	Real	--	--	N/A	m	7.3.3

[3] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

7.3.1 Attribute 33 – x

x, *y*, and *z* are the WGS84 coordinates using a Cartesian, Earth-Centered, Earth-Fixed (ECEF) coordinate system. Figure 21 illustrates the earth's ellipsoid with the ECEF origin (0, 0, 0) and axes in green. The ECEF origin is the earth's center of mass. The ECEF x-axis points to the intersection of the equator and the IERS Reference Meridian (IRM) (Greenwich meridian), the z-axis points towards the North Pole (true North), and the y-axis completes the right-handed coordinate system. An example ECEF position shows its x, y, and z coordinates. All ECEF measurements use the International System of Units (SI) of meters.

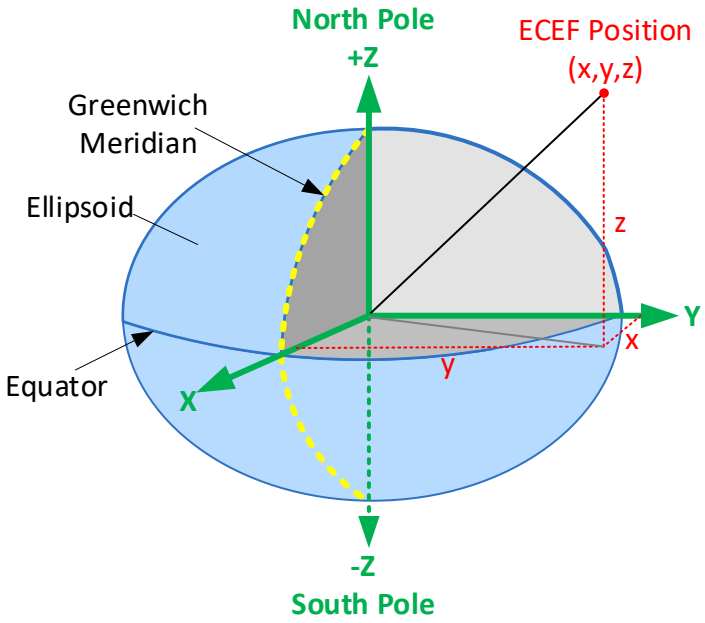


Figure 21: ECEF Coordinate System

7.3.2 Attribute 34 – y

Refer to Attribute 33 for the details of this attribute.

7.3.3 Attribute 35 – z

Refer to Attribute 33 for the details of this attribute.

7.4 AbsGeodetic Class

The AbsGeodetic class has attributes for an Absolute Geodetic position.

Table 6: AbsGeodetic Class Attributes (extends Position)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	lat	Real	-HALF_PI	HALF_PI	N/A	rad	7.4.1
34	lon	Real	-PI	PI	N/A	rad	7.4.2
35	hae	Real	-12 000.0	--	N/A	m	7.4.3
36	confidence	Confidence	N/A	N/A	N/A	None	7.4.4

7.4.1 Attribute 33 – lat

Latitude (*lat*), Longitude (*lon*), and Height-Above-Ellipsoid (*hae*) are the WGS84 coordinates using a geodetic coordinate system. Figure 22 illustrates a geodetic position above the earth's ellipsoid. The zero longitude is at the intersection of the equator and the IERS Reference Meridian (IRM) (Greenwich meridian), the zero latitude at the equator. An example geodetic position shows its lat, lon, and hae coordinates. The minimum value is set to below the lowest point on the earth, Challenger Deep, at the bottom of the Mariana Trench.

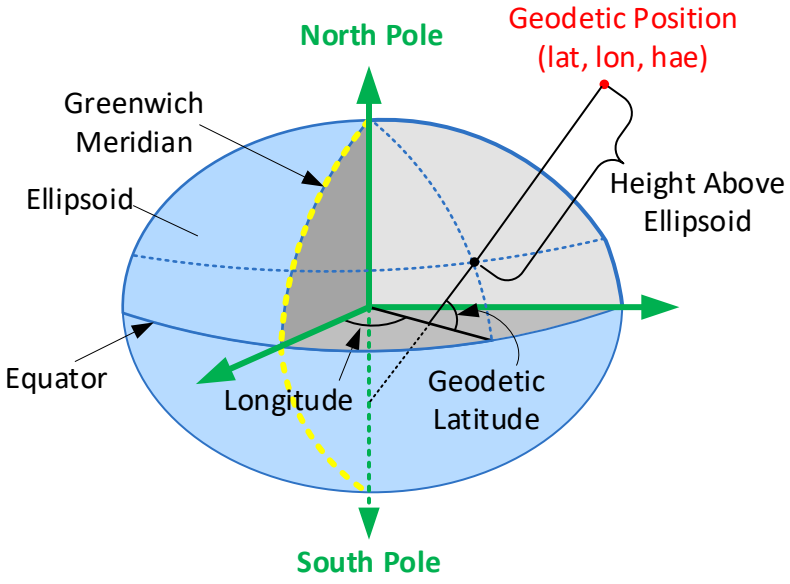


Figure 22: Geodetic Coordinate System

7.4.2 Attribute 34 – lon

Refer to the description of Attribute 33 for the details of this attribute.

7.4.3 Attribute 35 – hae

Refer to the description of Attribute 33 for the details of this attribute.

7.4.4 Attribute 36 – confidence

This attribute is a composite relationship to the Class Confidence. Section 7.5 defines the Confidence class.

7.5 Confidence Class

The Confidence class provides the computed Circular Error and Linear Error at a confidence level (typically 50% or 90%) for the given geodetic coordinate. For example, CE90/LE90 have circularError and linearError values set with the level set to 90. The Confidence class attributes are different than the Base Attributes standard deviation and cross correlation values. Table 7 lists the attributes for the Confidence Class.

Table 7: Confidence Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	circularError	Real	0.0	--	N/A	m	7.5.1
34	linearError	Real	0.0	--	N/A	m	7.5.2
35	level	Real	0.0	100.0	N/A	%	7.5.3

To properly use and interpret the errors the confidence level must be part of the instance of the class.

Requirement	
ST 1906.1-05	When using a Confidence class instance and the circularError or linearError attributes are defined, the producer shall include the level attribute.

7.5.1 Attribute 33 – circularError

The *circularError* is the error distance on the ellipsoid plane (see Figure 7) at the given geodetic point, for the given confidence level (level).

7.5.2 Attribute 34 – linearError

The *linearError* is the vertical error distance orthogonal to the ellipsoid plane (see Figure 7) at the given geodetic point, for the given confidence level (level).

7.5.3 Attribute 35 – level

The *level* is the error confidence value.

7.6 RelPosition Class

The RelPosition class has attributes for specifying a relative position in a Cartesian coordinate system. When using an RelPosition instance, all three attributes, x, y, and z, need to have numerical values and not be Unknown as defined by the Motion Imagery Handbook for Report-On-Change. Should one of the values become Unknown then all three values are Unknown.

Requirement	
ST 1906.2-07	When using an RelPosition instance, and one or more of the x, y, or z attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown.

Table 8: RelPosition Class Attributes (extends Position)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	x	Real	--	--	N/A	m	7.6.1
34	y	Real	--	--	N/A	m	7.6.2
35	z	Real	--	--	N/A	m	7.6.3

7.6.1 Attribute 33 – x

x, *y*, and *z* are relative Cartesian coordinates from the parent stage's coordinate system. If the stage is the root stage, i.e., parentStage is zero, the coordinates define a relative constellation of points with the origin at -x, -y, and -z. This is useful for cases where the absolute position of a Motion Imagery System is unknown.

7.6.2 Attribute 34 – y

Refer to the description of Attribute 33 for the details of this attribute.

7.6.3 Attribute 35 – z

Refer to the description of Attribute 33 for the details of this attribute.

7.7 Orientation Class

The Orientation class contains attributes for the orientation associated with a Position.

Table 9: Orientation Class Attributes[4]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	absEnu	AbsEnu	N/A	N/A	N/A	None	7.7.1
34	relOrientation	RelOrientation	N/A	N/A	N/A	None	7.7.2
35	reserved	RESERVED	N/A	N/A	N/A	None	7.7.3
36	kinematics	LIST<Kinematic>	--	--	N/A	None	7.7.4
37	magneticHeading	Real	0.0	TWO_PI	N/A	rad	7.7.5
38	course	Real	0.0	TWO_PI	N/A	rad	7.7.6
39	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.7.7
40	absEcef	AbsEcef	N/A	N/A	N/A	None	7.7.8

[4] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables.

7.7.1 Attribute 33 – absEnu

This attribute is a specialization relationship to the Orientation class. Section 7.8 defines the AbsEnu class.

7.7.2 Attribute 34 – relOrientation

This attribute is a specialization relationship to the Orientation class. Section 7.9 defines the RelOrientation class.

7.7.3 Attribute 35 – Reserved

This class reserves an attribute identifier (*Reserved*) for a future alternate method of specifying orientations using quaternions.

7.7.4 Attribute 36 – kinematics

This attribute is a list of Kinematic classes. Section 7.10 defines the Kinematic class.

7.7.5 Attribute 37 – magneticHeading

The *magneticHeading* attribute is the angle the stage's reference vector is pointing relative to magnetic north. The ENU-LRP reference vector is the projection of the stage's x-axis onto the ENU-LRP. Figure 23 shows an aircraft with its ENU-LRP below the aircraft for illustrative purposes. The aircraft pitch is upward, and the heading is in a north-easterly direction. The illustration shows the aircraft stage's x-axis projection onto the ENU-LRP, providing the heading vector. The angular measure, in radians, from Magnetic North to the heading vector is the magnetic heading.

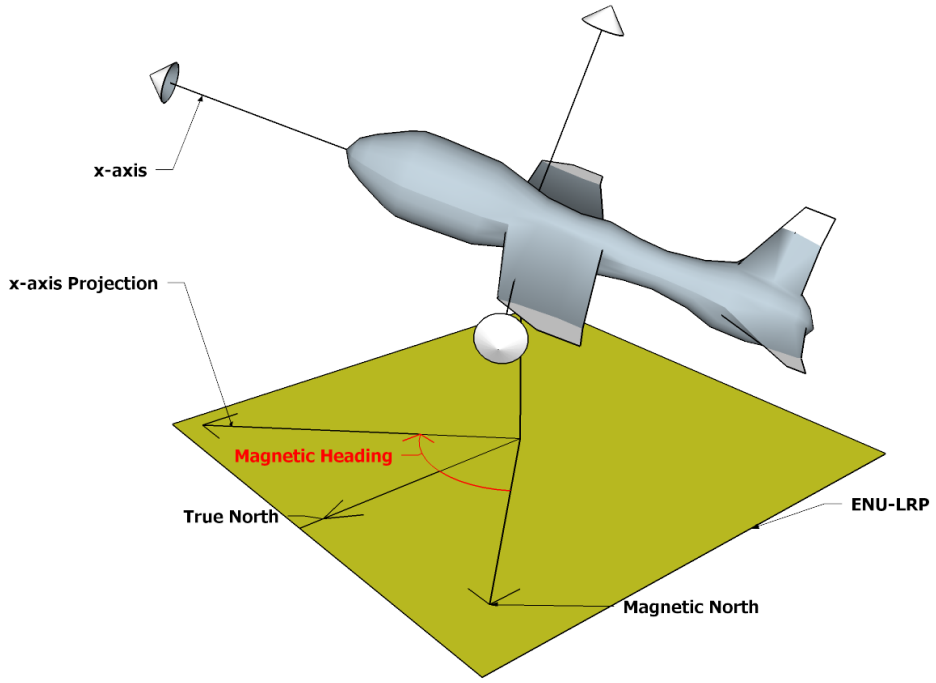


Figure 23: Illustration of Magnetic Heading

If the stage's x-axis is orthogonal to the ENU-LRP (i.e., straight up or down), the magnetic heading is undefined.

7.7.6 Attribute 38 – course

The *course* attribute is the measure of the direction, in radians, the stage is moving (not necessarily the direction the stage's x-axis is pointing). Figure 24 illustrates the course for a platform stage in red; the platform's x-axis is pointing southwest; the wind is from the west-northwest (to the east-southeast) – the platform's "course" is moving to the southeast. Course is directly measurable by on-board navigation or estimated computationally by comparing the last known position to the current position.

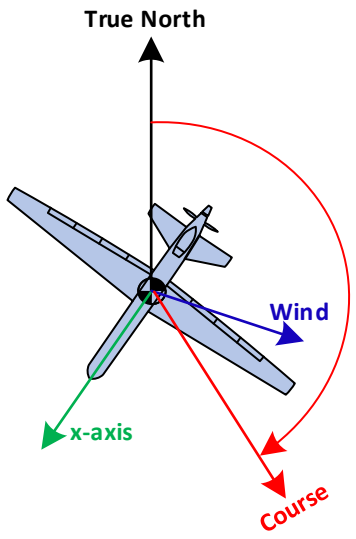


Figure 24: Platform course compared to other directional data.

7.7.7 Attribute 39 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.7.8 Attribute 40 – absEcef

This attribute is a specialization relationship to the Orientation class. Section 7.13 defines the AbsEcef class.

7.8 AbsEnu Class

The AbsEnu class has attributes to specify orientations for a Position using absolute East-North-Up (ENU) frame of reference.

Table 10: AbsEnu Class Attributes (extends Orientation)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	rotAboutEast	Real	-PI	PI	N/A	rad	7.8.1
34	rotAboutNorth	Real	-PI	PI	N/A	rad	7.8.2
35	rotAboutUp	Real	-PI	PI	N/A	rad	7.8.3

7.8.1 Attribute 33 – rotAboutEast

The ENU orientation has the X, Y, and Z axes aligned with the East, North, and Up respectively. For any given absolute position on, above, or below the WGS84 ellipsoid (except at the center of the earth), the position defines the origin of a local reference coordinate system. As Figure 25 illustrates, the orientation of the coordinate system has the X-Y plane (yellow) parallel to a plane (blue) that is tangent to the ellipsoid; where the X-axis aligns with the latitude or meridian lines,

with positive values going east; the Y-axis aligns with longitude lines, with positive values going north. The Z-axis is orthogonal to the planes (i.e., geodetically) with positive values increasing in the upwards direction, away from the North-South pole, or WGS84 Z-axis, (unless the position is at either pole, then the positive direction is away from the earth's center of gravity). With some platforms and sensors, heading, elevation, and roll angles may use different ranges (i.e., heading uses 0 to TWO_PI degrees), so range adjustments may be necessary to map to -PI to PI. When the *rotAboutEast*, *rotAboutNorth*, or *rotAboutUp* attribute values become ROC_Unknown they default to zero.

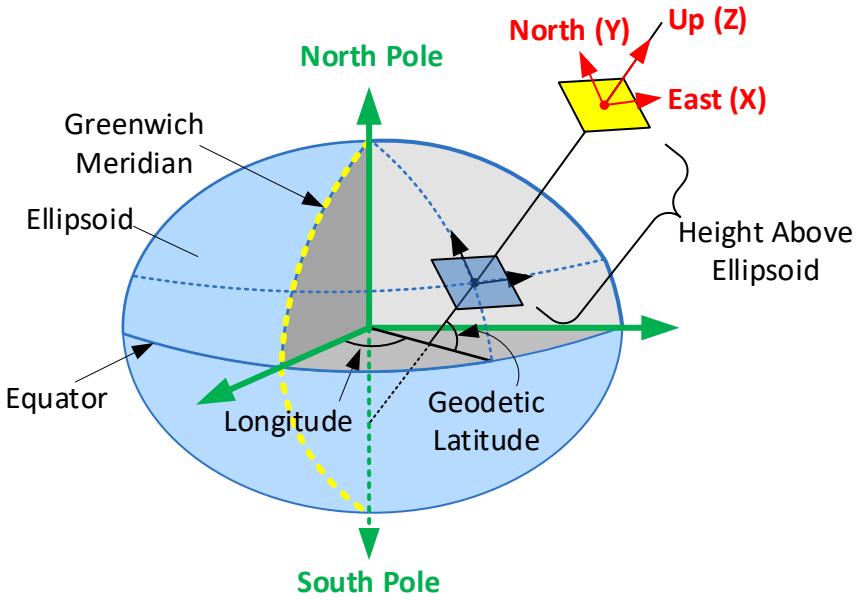


Figure 25: East-North-Up (ENU) Orientation

7.8.2 Attribute 34 – rotAboutNorth

Refer to the description of Attribute 33 for the details of this attribute.

7.8.3 Attribute 35 – rotAboutUp

Refer to the description of Attribute 33 for the details of this attribute.

7.9 RelOrientation Class

The RelOrientation class has attributes specifying a stage's relative orientation to the parent's stage.

Table 11: RelOrientation Class Attributes (extends Orientation)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	alpha	Real	--	--	N/A	rad	7.9.1
34	beta	Real	--	--	N/A	rad	7.9.2
35	psi	Real	--	--	N/A	rad	7.9.3

7.9.1 Attribute 33 – alpha

alpha, *beta*, and *psi* are relative orientations to the parent stage's coordinate system. If the stage is the root stage, i.e., parentStage is zero, the orientations define a relative constellation of points with a reference orientation of origin -alpha, -beta, and -psi. This is useful for cases where the absolute orientation of a Motion Imagery System is unknown.

7.9.2 Attribute 34 – beta

Refer to the description of Attribute 33 for the details of this attribute.

7.9.3 Attribute 35 – psi

Refer to the description of Attribute 33 for the details of this attribute.

7.10 Kinematic Class

The Kinematics class list enables the addition of kinematic information (velocity, acceleration, etc.) for Position and Orientation information.

Table 12: Kinematic Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	k1	Real	--	--	N/A	See Description	7.10.1
34	k2	Real	--	--	N/A	See Description	7.10.2
35	k3	Real	--	--	N/A	See Description	7.10.3
36	reserved	RESERVED	N/A	N/A	N/A	None	7.10.4
37	level	UInt	1	--	N/A	None	7.10.5
38	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.10.6

7.10.1 Attribute 33 – k1

Each sub-class of the Position class (i.e., AbsGeocentric, AbsGeodetic, RelPosition) and Orientation class (i.e., AbsEnu, RelOrientation) provides three attributes that relate to the three kinematic *k1*, *k2*, and *k3* attributes. Table 13 defines the relationship between the Kinematic class's attributes and the attributes of these classes.

Table 13: Relationship between Kinematics attributes to Position-based attributes

	Positions			Orientations	
Kinematics	**AbsGeocentric**	**AbsGeodetic**	**RelPosition**	**AbsEnu**	**AbsEcef or RelOrientation**
k1	x	lat	x	rotAboutEast	alpha
k2	y	lon	y	rotAboutNorth	beta
k3	z	hae	z	rotAboutUp	psi

For example, an AbsGeodetic position defines lat, lon, and hae values, per Table 13, the kinematics values k1, k2, and k3 relate to the lat, lon, and hae respectively. With the level

attribute set to 1, the k1, k2, and k3 values will be the velocities for their respective Position value, i.e., the value k1 is the velocity in the latitude direction, k2 is the velocity in the longitude direction and k3 is the velocity in the hae direction. The units for the velocity k-values are the positions units/sec.

The k-values are specific time derivatives of the attributes for the specific level value. The first level or derivative of the attribute is velocity, the second derivative is acceleration, the third derivative is jerk, etc. Every Position or Orientation instance has the potential to include velocity, acceleration, jerk, snap, crackle, pop, and other higher-level derivatives if needed. The units for the k-values are the attribute's unit (e.g., meters) divided by the *level* power of seconds, or $\frac{units}{seconds^{level}}$, for example, $\frac{meters}{second}$ for a velocity or $\frac{meters}{second^2}$ for an AbsGeocentric acceleration.

7.10.2 Attribute 34 – k2

Refer to the description of Attribute 33 for the details of this attribute.

7.10.3 Attribute 35 – k3

Refer to the description of Attribute 33 for the details of this attribute.

7.10.4 Attribute 36 – reserved

This class reserves an attribute identifier (*Reserved*) for a fourth value (k4) in support of a future update to include quaternions.

7.10.5 Attribute 37 – level

Refer to the description of Attribute 33 for the details of this attribute.

7.10.6 Attribute 38 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.11 Source Class

The Source class attributes specify information about how the position measurement was determined.

Table 14: Source Class Attributes[5]

Id	Name	Type	Min	Max	MLen	Units	Ref
33	positionMethod	PositionMethod	N/A	N/A	N/A	None	7.11.1
34	numNavSatInView	UInt	--	--	N/A	count	7.11.2

7.11.1 Attribute 33 – positionMethod

This attribute is an enumeration. Section 8.1 defines the PositionMethod enumeration.

7.11.2 Attribute 34 – numNavSatInView

The *numNavSatInView* attribute is the number of satellites a positioning system (e.g., GPS receiver) uses to determine the position.

7.12 RelVector Class

The RelVector (Relative Vector) class provides attributes to specify a relative position using two angles (azimuth, elevation) and a range. The RelVector is an equivalent but alternate method for specifying the RelPosition class information, see Section 6.1.1 for further information. When using an RelVector instance, all three attributes, azimuth, elevation, and range, need to have numerical values and not be Unknown as defined by the Motion Imagery Handbook for Report-On-Change. Should one of the values become Unknown then all three values are Unknown.

Requirement	
ST 1906.2-10	When using an AbsEnu instance, and one or more of the azimuth, elevation, or range attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown.

Table 15: RelVector Class Attributes (extends Position)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	azimuth	Real	--	--	N/A	rad	7.12.1
34	elevation	Real	--	--	N/A	rad	7.12.2
35	range	Real	0.0	--	N/A	m	7.12.3

[5] For consistency across the MIMD suite of documents, MISB ST 1901 defines the label designations for the columns in the attribute tables. Capitalized attribute table items (e.g., PI) are constant values listed in MISB ST 1901.

7.12.1 Attribute 33 – azimuth

The combination of azimuth, elevation, and range defines a vector relative to the parent stage's origin to the origin of this stage. The azimuth is an angle measurement from the parent stage's x-axis around the parent's z-axis toward the direction of the vector. The elevation is an angle measurement from the parent stage's x-y plane toward the direction of the vector. The range is the distance in meters from the parent stage's origin. See Section 6.1.1.2 for an illustration and more information on relative angular vectors.

7.12.2 Attribute 34 – elevation

Refer to the description of Attribute 33 for the details of this attribute.

7.12.3 Attribute 35 – range

Refer to the description of Attribute 33 for the details of this attribute.

7.13 AbsEcef Class

The AbsEcef class attributes specify orientations using absolute the Earth-Centered-Earth-Fixed (ECEF) frame of reference.

Table 16: AbsEcef Class Attributes (extends Orientation)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	alpha	Real	-PI	PI	N/A	rad	7.13.1
34	beta	Real	-PI	PI	N/A	rad	7.13.2
35	psi	Real	-PI	PI	N/A	rad	7.13.3

7.13.1 Attribute 33 – alpha

Attributes *alpha*, *beta*, and *psi* are orientations relative to the ECEF coordinate system. Figure 26 illustrates the ECEF axis and a point with Absolute Geocentric Orientation. The figure shows the ECEF's positive Z axis is from the center of the along the rotational axis, toward the north pole; the positive X axis is from the center of the earth to the where the Greenwich meridian and equator intersect; and the Y axis is the right-handed orthogon axis to the Z and X axes.

The figure illustrates the Absolute Geocentric Orientation at the point in the middle of the yellow plane. The dashed black line shows the point's location relative to the earth (with X, Y, Z coordinates not shown). The yellow plane's green axes align with the earth's ECEF axes. The yellow plane's red axes are the offsets from the ECEF axes, specifically α is the rotation around the Z axis and ψ is the rotation around the X axis, and there is not a rotation around the Y axis, so $\beta = 0$.

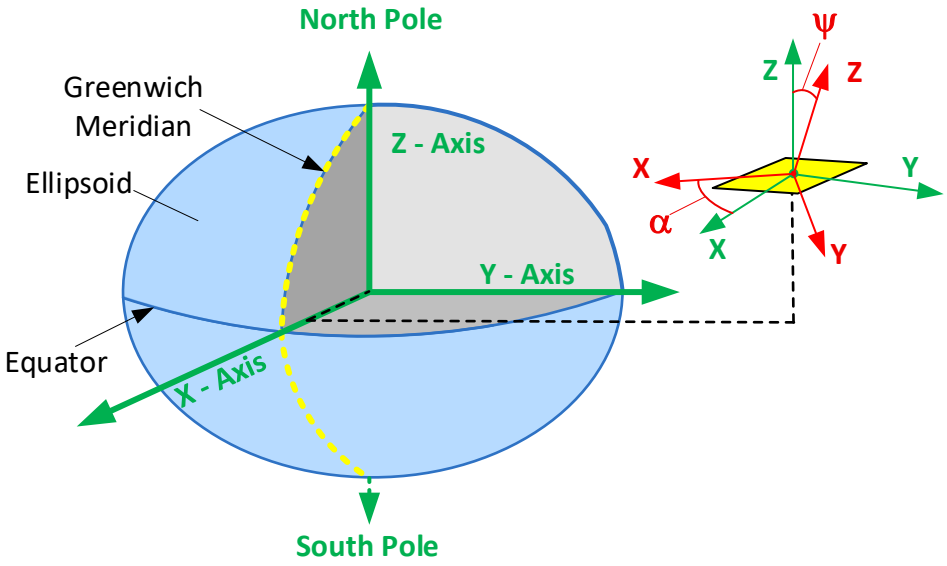


Figure 26: East-North-Up (ENU) Orientation

7.13.2 Attribute 34 – beta

Refer to the description of Attribute 33 for the details of this attribute.

7.13.3 Attribute 35 – psi

Refer to the description of Attribute 33 for the details of this attribute.

8 Model Enumerations

8.1 PositionMethod Enumeration

The *positionMethod* attribute is a value from the enumeration list in Table 17.

Table 17: Enumeration Values for PositionMethod

Id	Name	Description
0	INS	On-board Inertial Navigation System
1	NAVSAT_GPS	GPS - U.S.
2	NAVSAT_GALILEO	Galileo - E.U.
3	NAVSAT_QZSS	QZSS - Japan
4	NAVSAT_NAVIC	NAVIC - India
5	NAVSAT_GLONASS	GLONASS - Russia
6	NAVSAT_BEIDOU_1	BeiDou-1 - China
7	NAVSAT_BEIDOU_2	BeiDou-2 - China
8	FIXED	Fixed point on earth

Appendix A Examples

A.1 Single Point Sensor

An example of a Single Point Sensor is a camera with built-in GPS and orientation measurements. Figure 27 shows an illustration of a camera and its X, Y, Z axis. The position of the camera is the origin of the three axis.

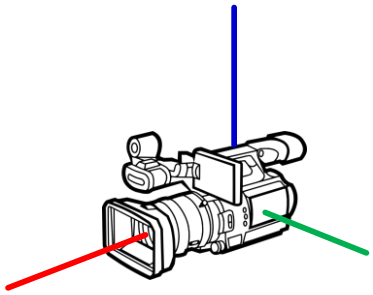


Figure 27: Single Point Sensor

Using a geodetic position of the origin and a position relative to ENU, the staging class instances in an outline format are:

Stage 1

parentStage = 0

Position

lat = *value*

lon = *value*

hae = *value*

Orientation

rotAboutEast = *value*

rotAboutNorth = *value*

rotAboutUp = *value*

Where *value* is a numerical value within the range specified by the attributes. The parentStage = 0 value indicates the stage is the root stage.

A.2 Airborne System

Figure 28 is an illustration of stages on-board a UAS. The root stage is the location provided by the GPS system. The second stage is the relative position and orientation of the UAS center-of-gravity (CG) to the GPS. The third (and terminal) stage is the relative position of the sensor to the CG and its orientation.

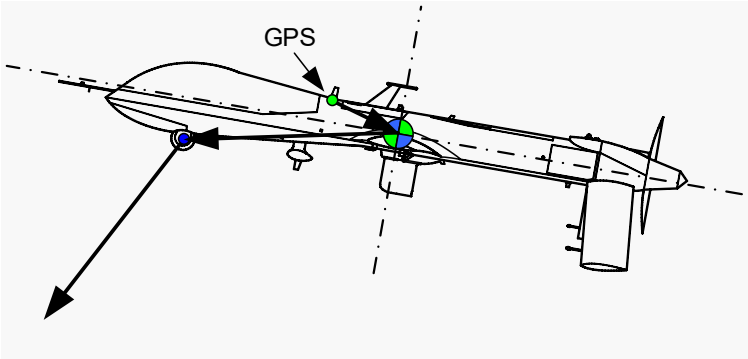


Figure 28: Airborne System

Table 18 lists the stages for this example.

Table 18: Airborne System Staging List Example

Stage 1 (GPS)	Stage 2 (CG)	Stage 3 (Sensor)
mimdId = 12 parentStage = 0 Position lat = *value* lon = *value* hae = *value*	mimdId = 15 parentStage = 12 Position x = *value* y = *value* z = *value* Orientation rotAboutEast = *roll* rotAboutNorth = pitch rotAboutUp = *yaw*	mimdId = 7 parentStage = 15 Position x = *value* y = *value* z = *value* Orientation alpha = *value* beta = *value* psi = *value*

Depending on the relative distance between Stage 1 and Stage 2, the Stage 2 rotations may be adjusted to account for a different ENU reference plane.

For this example, the sensor is at the center of rotation of the ball turret; otherwise, there would be a fourth translational stage from the gimbal center of rotation to the sensor location.

A.3 Absolute Child to Relative Parent

Another example staging system based on Figure 28 is a sensor which contains its own GPS/INS system. In this case the sensor stage may still reference the CG stage (via the parentStage) but the sensor stage has its own absolute location and rotation. Table 19 lists the stages for this example.

Table 19: Airborne System Staging List Example

Stage 1 (GPS)	Stage 2 (CG)	Stage 3 (Sensor - Relative)	Stage 4 (Sensor - Absolute)
mimdId = 12 parentStage = 0	mimdId = 15 parentStage = 12	mimdId = 7 parentStage = 15	mimdId = 25 parentStage = 7

Position lat = *value* lon = *value* hae = *value*	Position x = *value* y = *value* z = *value* Orientation rotAboutEast = *pitch* rotAboutNorth = *roll* rotAboutUp = *yaw*	Position x = *value* y = *value* z = *value* Orientation alpha = *value* beta = *value* psi = *value*	Position lat = *value* lon = *value* hae = *value* Orientation rotAboutEast = *value* rotAboutNorth = *value* rotAboutUp = *value*

Since Stage 4 has a parent identifier linking it to Stage 3, and Stage 4 does not provide any relative information, the Position of Stage 3 and Stage 4 are the same, just measured differently (and therefore would have different uncertainties).

Appendix B Coordinate Conversions

The following equations are from DRAFT Objective SENSRB Mapping Description Document.

B.1 Geodetic-To-Geocentric Coordinate Transformation

WGS84 defines the values for the Earth model using the semi-major axis, ***a***, and the flattening, ***f***. These are exact numbers.

$$a = 6378137.0$$

$$f = \frac{1}{298.257223563} \quad \textbf{Equation 20}$$

Using the flattening and major axis value, implementers can calculate the values for eccentric and normal for the curvature of the earth. The geodetic to geocentric coordinate transformation requires eccentric and normal values along with latitude (ϕ). Equation 21 and Equation 22 describe this transformation.

$$e = \sqrt{(2f - f^2)} \quad \textbf{Equation 21}$$

$$N = \frac{a}{\sqrt{(1 - (e\sin(\phi))^2)}} \quad \textbf{Equation 22}$$

The conversion from geodetic to geocentric coordinates is a straightforward conversion opposed to the reverse iterative process. Given the latitude (ϕ), longitude (λ), and height above ellipsoid (HAE) the point of interest can converts to geocentric coordinates using Equation 23 through Equation 25.

$$X = (N + HAE)\cos(\phi)\cos(\lambda) \quad \textbf{Equation 23}$$

$$Y = (N + HAE)\cos(\phi)\sin(\lambda) \quad \textbf{Equation 24}$$

$$Z = (N(1 - e^2) + HAE)\sin(\phi) \quad \textbf{Equation 25}$$

The values resulting from Equation 23 through Equation 25 are the geocentric XYZ values in the ECEF frame.

B.2 Geocentric-To-Geodetic Coordinate Transformation

The defined values for the mathematical surface used to model the Earth (given below and considered exact) are the semi-major axis, ***a***, and the flattening, ***f***.

$$a = 6378137.0$$

$$f = \frac{1}{298.257223563} \quad \text{Equation 26}$$

Using the flattening and major axis value, Equation 27 and Equation 29 compute the values for eccentric and normal for the curvature of the earth. The geocentric to geodetic coordinate transformation requires these two values combined with the ECEF coordinates.

$$e = \sqrt{(2f - f^2)} \quad \text{Equation 27}$$

$$d = \sqrt{(X^2 + Y^2)} \quad \text{Equation 28}$$

$$\phi_1 = \frac{Z}{\sqrt{\left((1 - (e)^2) * \mathrm{d}\right)}} \quad \text{Equation 29}$$

The conversion from geocentric to geodetic coordinates is an iterative process. Given the ECEF position ($X, Y, and\ Z$) the point of interest can be converted to geodetic coordinates once the variable p in Equation 33 reaches an acceptable threshold.

$$N_1 = \frac{a}{\sqrt{(1 - (e \sin(\phi_1))^2)}} \quad \text{Equation 30}$$

$$h_1 = (N + HAE)\cos(\phi)\sin(\lambda) \quad \text{Equation 31}$$

$$\phi_2 = \operatorname{atan}\left(\left(\frac{Z}{d}\right) * \left(1 + \frac{\left((e^2) * N_1 * sin(\phi_1)\right)}{Z}\right)\right) \quad \text{Equation 32}$$

$$p = abs(\phi_2 - \phi_1) \quad \text{Equation 33}$$

After the iterative process has converged the latitude (ϕ), longitude (λ), and height above ellipsoid (HAE) are:

$$\phi = \phi_2 \quad \text{Equation 34}$$

$$\lambda = atan2(Y, X) \quad \text{Equation 35}$$

$$HAE = h_1 \quad \text{Equation 36}$$

Appendix C Rotations and Matrices

When documenting rotations for coordinate systems there are several intermixed factors to consider in creating matrices and using matrices to perform the operations.

C.1 Creating A Rotation Matrix

When creating a rotation matrix there are four options two choose from: Left vs Right-handed, Tait-Bryan vs Euler, the Primary Rotation Order, and Intrinsic vs Extrinsic.

1) **Left-handed vs. Right-handed system**: The rotations in the Staging System (and MIMD Model) use right-handed rotations. With right-handed rotations the following table describes the positive angular rotational movement around each axis:

Rotation About axis…	Goes from axis…	Toward axis…
X	+Y	+Z
Y	+Z	+X
Z	+X	+Y

2) **Tait-Bryan vs. Euler:** Both methods define rotations around axes. The Tait-Bryan method defines rotations around the three axes, x, y, and z. The Euler method defines rotations around two axes. With the Euler method one of the axes has a rotation repeated because rotations around one axis move the other two axes, and by the third rotation the original axis is in a different orientation.

3) **Primary Rotation Order:** The order of rotation around each axis is not commutative (i.e., $R_x * R_y \neq R_y * R_x$), so defining the order is important. For example, rotating around axis x first, then y, then z produces a different overall orientation than rotating around the y axis, then x axis, then z axis. Equation 37, Equation 38, and Equation 39 define the elementary rotations around the X, Y, or Z axes, respectively.

$$R_X(\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & -sin\,\psi \\ 0 & \sin\psi & \cos\psi \end{bmatrix} \quad \text{Equation 37}$$

$$R_Y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \quad \text{Equation 38}$$

$$R_Z(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Equation 39}$$

Where, ψ, β, α are the angles of rotations around X, Y, and Z, respectively. Combining these elementary rotations using matrix multiplication form a single rotation matrix $\widehat{R}$. For example, defining a rotation matrix $\widehat{R}$ as: $\widehat{R} = R_X(\psi) * R_Y(\beta) * R_z(\alpha)$ or $\widehat{R} = R_Y(\beta) * R_X(\psi) * R_z(\alpha)$. The Primary Rotation Order defines the axis rotation order, but the matrix multiplication order may reverse when using Intrinsic vice Extrinsic rotations.

4) **Intrinsic vs. Extrinsic:** When applying rotations there are two methods for performing the rotations: intrinsic or extrinsic.

   a. Intrinsic rotations apply each rotation to axes within the rotating coordinate system. Therefore, the first rotation moves the second and third axis of rotation, the second rotation is around the "new" second axis, and the third rotation is around the "new-new" third axis. Figure 29 illustrates an example where there are three rotations with Primary Rotation Order of X, Y, and then Z, and rotation angles of 60, 45, and 30 degrees, respectively.

The first rotation in figure (a) is a 60-degree rotation around the x axis, which rotates the y axis upwards (and the z axis downwards) from their original positions. The rotation results in two new axes, New-Y and New-Z, shown in red. With **intrinsic** rotation, the second rotation of 45 degrees, in figure (b), is around the new rotated y axis, "New-Y". The rotation repositions the original x axis and the New-Z axis. The second rotation results in new axis locations New-X and New-New Z, shown in green. The third 30-degree rotation, in figure (c), is around the New-New Z axis, resulting in the blue axis.

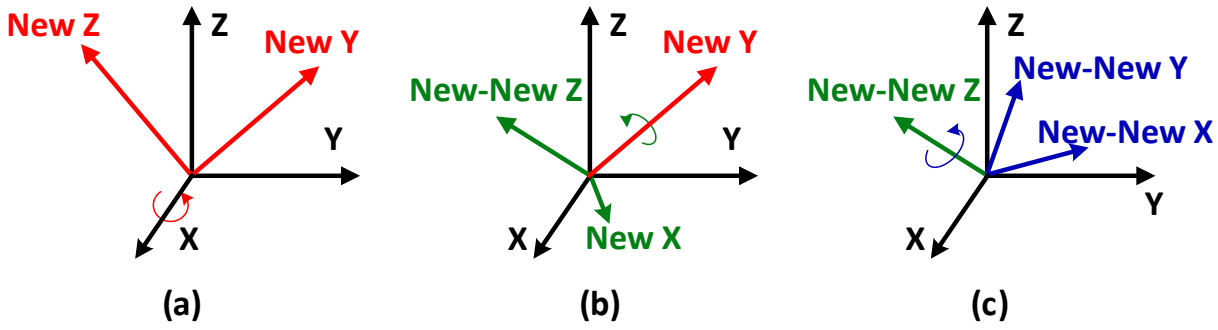


Figure 29: Intrinsic Rotations for Primary Rotation Order X, Y, and Z

With intrinsic rotations the rotation matrix multiplies its axis-rotations in order of the Primary Rotation Order. For example, if the Primary Rotation Order is X, Y, and Z, the intrinsic rotation matrix is $\widehat{R} = R_X(\psi) * R_Y(\beta) * R_Z(\alpha)$.

b. Extrinsic rotations apply each rotation to the original axis positions as if they have not moved. Figure 30 illustrates an example where there are three rotations with Primary Rotation Order of X, Y, and then Z, and rotations angles of 60, 45, and 30 degrees, respectively. The first rotation in figure (d) is a 60-degree rotation around the x axis, which rotates the y axis upwards (and the z axis downwards) from their original positions.

   The first rotation results in two new axes, New-Y and New-Z, shown in red. The first rotation is the same as the intrinsic method; however, with **extrinsic** rotations the second rotation, figure (e), of 45-degrees is around the original y axis (not the newly rotated y axis). The rotation results in three new axes, New-X, New-New-Y and New-New-Z, shown in green. The third 30-degree rotation, in figure (c), is around the original z axis, resulting in the blue axis.

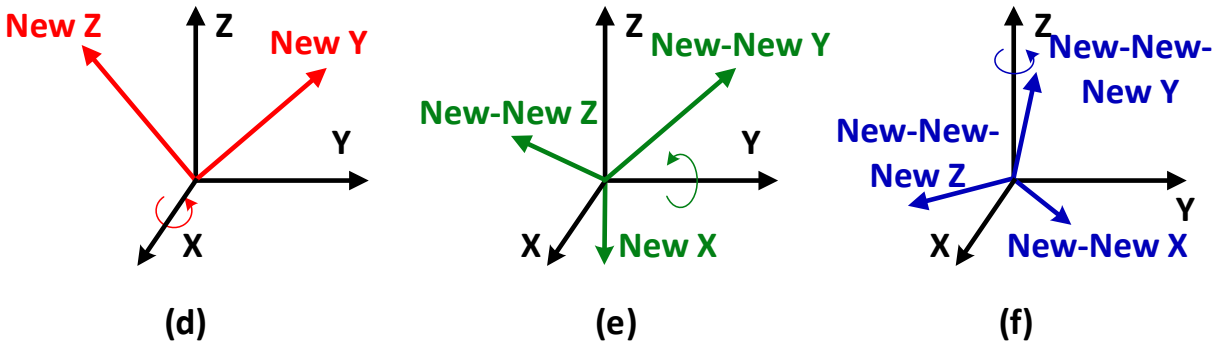


Figure 30: Extrinsic Rotations for Primary Rotation Order X, Y

With extrinsic rotations the rotation matrix multiplies is axis-rotation in reverse order of the Primary Rotation Order. For example, if the Primary Rotation Order is X, Y, and Z, the extrinsic rotation matrix is $\hat{R} = R_Z(\psi) * R_Y(\beta) * R_X(\alpha)$.

The Staging System's Rotation matrices are:
- **Right-handed,**
- **Tait-Bryan,**
- **ZYX Rotation Order,**
- **Intrinsic.**

C.2 Rotation Operations

Rotation matrices are unitary, i.e., their transpose is equal to their inverse.

Given: $R(\theta)$

$$R^{-1}(\theta) = R^T(\theta) = R(-\theta)$$

C.3 Rotation Matrix Application

The traditional interpretation of rotation matrices is they enable points or vectors in one coordinate system to either "rotate" within the coordinate system (Alibi) or "transfer" to a different coordinate system (Alias), depending on either left- or right-sided matrix multiplication.

An alternate interpretation of rotation matrices is to only consider them as transformations between two coordinate systems.

C.3.1 Alibi vs. Alias Interpretation

There are two methods for applying a rotation matrix to a vector: Alibi or Alias.

a. Alibi (also known as Active) rotation is where a vector rotates within a fixed coordinate system. In this writeup, when applying a rotation, $\hat{R}$, to a vector, $\vec{u}$, if the vector is post-multiplying (i.e., $\vec{v} = \hat{R} * \vec{u}$), the operation is Alibi. The resulting coordinates for the new vector after the Alibi rotation are for a vector which has rotated within the base coordinate system. Figure 31 illustrates an example of an Alibi rotation of 60 degrees

around the Z-axis which comes out of the page. The vector u (in red) lies on the x-axis with coordinates $(x, y) = (1, 0, 0)$, the alibi rotation produces vector v (in blue) using:

$$\vec{v} = R_Z(60) * \vec{u} = \begin{bmatrix} \cos 60 & -\sin 60 & 0 \\ \sin 60 & \cos 60 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.50 \\ 0.86 \\ 0 \end{bmatrix}$$

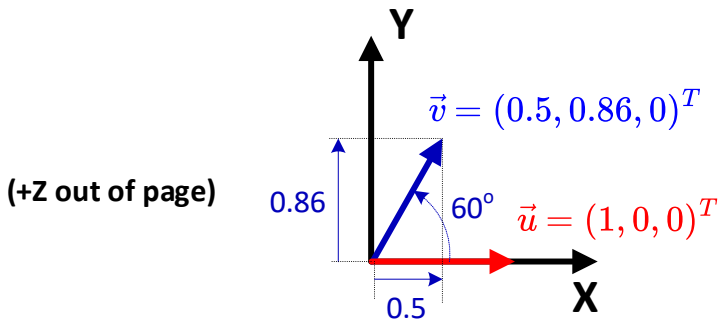


Figure 31: Alibi rotation Example.

b. Alias (also known as Passive) is where the coordinate system rotates, and the vector is "fixed" in space. When applying a rotation, $\hat{R}$, to a vector, $\vec{u}$, if the vector is pre-multiplying (i.e., $\vec{v} = \vec{u} * \hat{R}$), the operation is alias. The resulting coordinates are for the same vector (without moving) in the new coordinate system. Figure 32 illustrates an example of an alias rotation of 60 degrees. The figure (a) shows the original axis and the vector $\vec{u}$ in red before the rotation. The vector $\vec{u}$ has coordinates of $(x, y) = (1, 0)$. In figure (b) the alias rotation rotates the XY axis 60 degrees (green) making a new coordinate system with axis $X'$and Y'. The alias rotation equation produces the coordinates of the original unmoved vector (blue/red) but in the $X'$and Y' coordinate system's frame of reference; these coordinates are in vector $\vec{v}$ (blue):

$$\vec{v}^T = \vec{u}^T * R_Z(60) = [1 \quad 0 \quad 0] * \begin{bmatrix} \cos 60 & -\sin 60 & 0 \\ \sin 60 & \cos 60 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [0.5, -0.86 \quad 0]$$

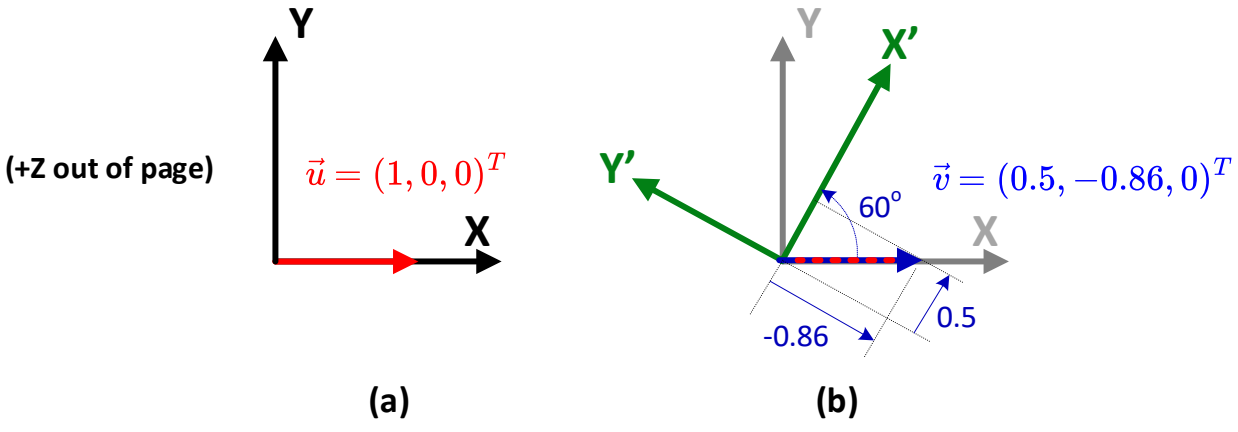


Figure 32: Alias rotation Example.

When applying rotations, $\hat{R}$, to a vector, $\vec{v}$, if the first axis of rotation (from the ordering) is next to the vector the rotation will be intrinsic. For example, if the Primary Rotation Order is X, Y, Z (i.e., $\hat{R} = R_X(\psi) * R_Y(\beta) * R_z(\alpha)$) and we are pre-multiplying, i.e., $\vec{v}_2 = \vec{v} * \hat{R}$, then because the vector is being multiplied by $R_X(\psi)$ first, the rotation is intrinsic. To have extrinsic rotations using the same Primary Rotation Order, define $\hat{R}_{extrinsic} = R_z(\alpha) * R_Y(\beta) * R_X(\psi)$ and pre-multiply $\vec{v}$, so $\vec{v}_2 = \vec{v} * \hat{R}_{extrinsic}$. This means one Primary Rotation Order's intrinsic matrix multiplication order, is a different Primary Rotation Order's extrinsic matrix multiplication order (and vice versa).

Rotations can become confusing since the order of the rotations can have different meanings depending on if they are intrinsic or extrinsic and depending on if the application of Alias or Alibi.

C.3.1.1 Alias and Alibi Relationship

An alternate way of interpreting the alias rotation is to recognize it is the negative alibi rotation. This equality shown mathematically is:

$$\vec{v} = R(\theta) * \vec{u}$$

$$\vec{v}^T = (R(\theta) * \vec{u})^T = \vec{u}^T * R(\theta)^T = \vec{u}^T * R(\theta)^{-1} = \vec{u}^T * R(-\theta)$$

This holds true for multiple axis rotations, i.e., $R(\alpha, \beta, \psi) * \vec{u} \equiv \vec{u}^T * R(-\alpha, -\beta, -\psi)$. Using this method of interpretation may cause confusion and is not recommended.

C.3.2 Transformation Interpretation

Mathematically, rotation matrices are a type of transformation from one coordinate system to another. Starting with a base Cartesian coordinate system, C_1, with axes X_1, Y_1, and Z_1, construct a second Cartesian coordinate system, C_2, sharing the same origin but with two or three of the orthogonal X_2, Y_2, and Z_2 axes pointing in different directions than C_1. Figure 33 illustrates coordinate system C_1 in black and coordinate system C_2 in blue, both sharing the same origin. The figure shows coordinate system C_2 "rotated" or oriented differently than C_1.

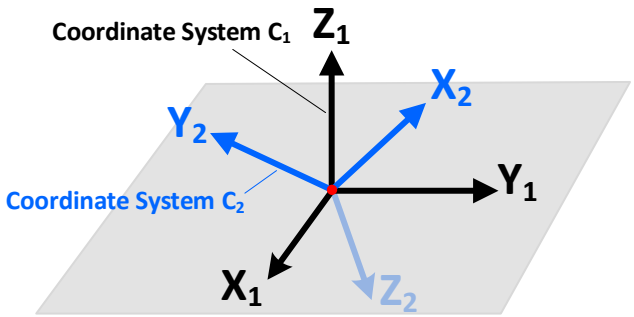


Figure 33: Two Cartesian Coordinate Systems C_1 and C_2, with C_2 oriented differently

A rotation matrix, R, defines the three-axis rotations (see Appendix C.1) to "rotate" from C_1 to C_2. The rotation matrix R is a transformation matrix, mapping vectors in C_1 to C_2 and vice versa.

To show the rotation matrix is a transformation from C_1 to C_2, the rotation notation is, ${}^{C_1 \leftrightarrow C_2}R$. The position of the C_1 on the left-hand side and C_2 on the right-hand side of the arrow is important when transforming vectors. Figure 34 illustrates the axis from Figure 33 separated into two distinct coordinate systems, along with the rotation matrix, ${}^{C_1 \leftrightarrow C_2}R$, which transforms vectors in C_1 to C_2 and vice versa.

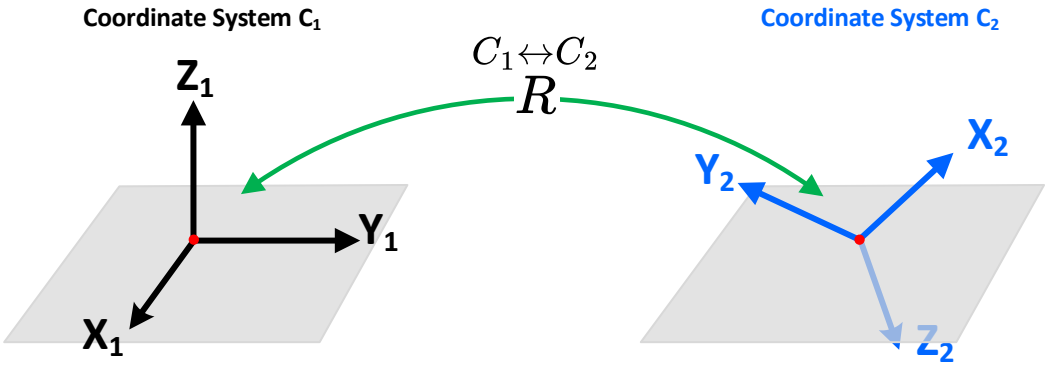


Figure 34: Transformation from C_1 to C_2 using ${}^{C_1 \leftrightarrow C_2}R$

To perform a $C_1 \rightarrow C_2$ transformation, multiply a vector, $\vec{v}_1$, (in C_1) on the left-hand side of ${}^{C_1 \leftrightarrow C_2}R$, resulting in a new vector, $\vec{v}_2$, in coordinate system C_2, i.e.,

$$\vec{v}_1 * {}^{C_1 \leftrightarrow C_2}R = \vec{v}_2$$

To perform a $C_1 \leftarrow C_2$ transformation, multiply a vector, $\vec{v}_2$, (in C_2) on the right-hand side of ${}^{C_1 \leftrightarrow C_2}R$, resulting in a new vector, $\vec{v}_1$, in coordinate system C_1, i.e.,

$$\vec{v}_1 = {}^{C_1 \leftrightarrow C_2}R * \vec{v}_2.$$

Figure 35 illustrates a simple rotation transformation example where coordinate system C_1 on the left has its X and Y axes aligned with the edges of the page and its Z axis coming out of the page. Coordinate system C_2 on the right is a rotation of the C_1 coordinate system around the Z_1 axis of 60 degrees, therefore ${}^{C_1 \leftrightarrow C_2}R = R_{Z_1}(60)$.

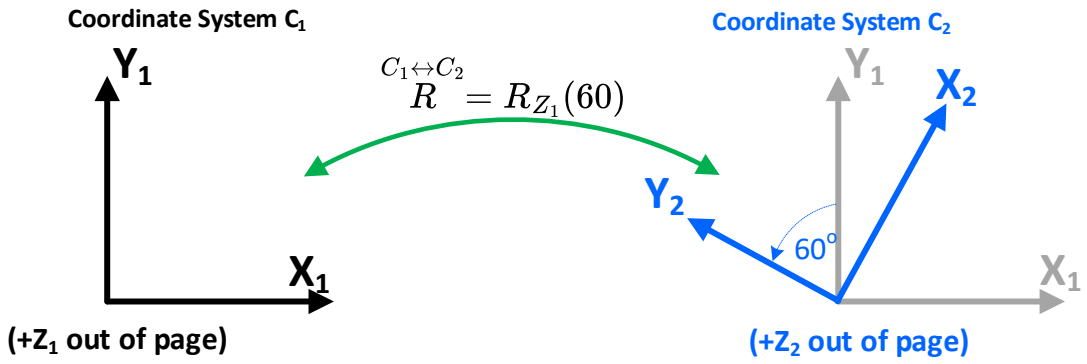


Figure 35: Rotation Transformation

Figure 36 illustrates an example of transformation of a vector, $\vec{u_1} = (1,0,0)^T$ in C_1 to C_2. The figure shows $\vec{u_1}$ in red in both coordinate systems. The transformation results in the coordinates of the vector in the C_2 coordinate system's frame of reference; these coordinates are the vector $\vec{u_2}$:

$$\overrightarrow{\boldsymbol{u_1}}^T * {}^{C_1\leftrightarrow C_2}_{\quad}R = [1, \quad 0, \quad 0] * \begin{bmatrix} \cos 60 & -\sin 60 & 0 \\ \sin 60 & \cos 60 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [0.5, -0.86, \quad 0] = \overrightarrow{\boldsymbol{u_2}}^T$$

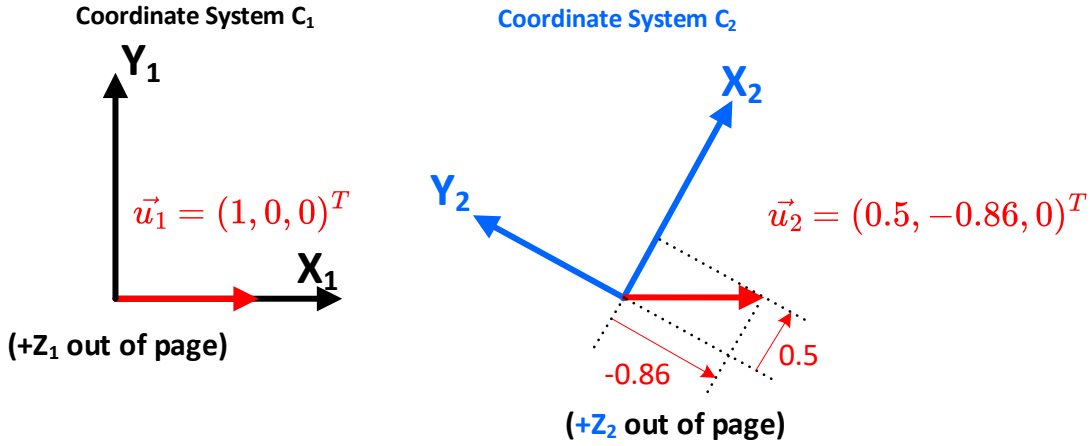


Figure 36: Transformation Example.

Figure 37 illustrates an example of transformation of a vector, $\vec{w_1} = (0,1,0)^T$ in C_2 to C_1. The figure shows $\vec{w_1}$ in pink in both coordinate systems. The transformation results in the coordinates of the vector in the C_1 coordinate system's frame of reference; these coordinates are the vector $\vec{w_2}$:

$${}^{C_1\leftrightarrow C_2}_{\quad}R * \overrightarrow{\boldsymbol{w_1}} = \begin{bmatrix} \cos 60 & -\sin 60 & 0 \\ \sin 60 & \cos 60 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = [-0.86 \quad 0.5 \quad 0] = \overrightarrow{\boldsymbol{w_2}}^T$$

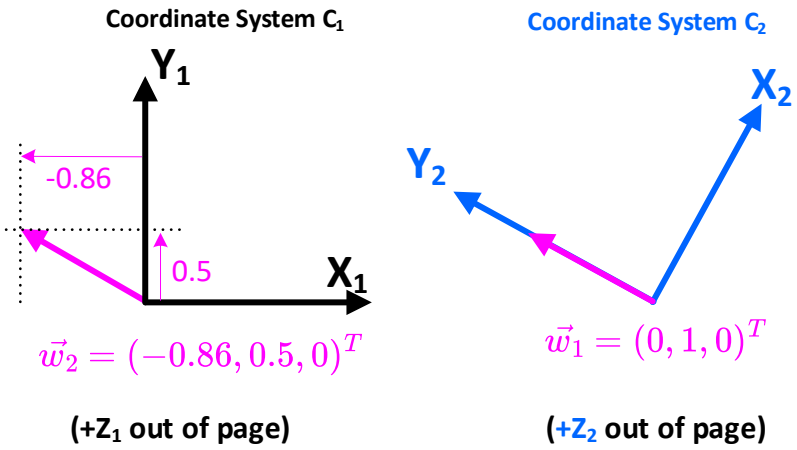


Figure 37: Transformation Example.

C.4 XYZ to ENU Rotation Matrix

To rotate the XYZ coordinate system to the ENU coordinate system the XYZ coordinate system rotates around the Z axis by the longitude angle plus 90 degrees $\left(\lambda_s + \frac{\pi}{2}\right)$ and then once around the new-X axis by 90 degrees minus the latitude angle $\left(\frac{\pi}{2} - \phi_s\right)$. Equation 40 defines the Rotation around Z, Equation 41 defines the rotation around the new-X axis.

$$R_Z(\lambda_s) = \begin{bmatrix} \cos(\lambda_s + 90) & -\sin(\lambda_s + 90) & 0 \\ \sin(\lambda_s + 90) & \cos(\lambda_s + 90) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\sin(\lambda_s) & -\cos(\lambda_s) & 0 \\ \cos(\lambda_s) & -\sin(\lambda_s) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{Equation 40}$$

$$R_X\left(\frac{\pi}{2} - \phi_s\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\frac{\pi}{2} - \phi_s & -sin\frac{\pi}{2} - \phi_s \\ 0 & \sin\frac{\pi}{2} - \phi_s & \cos\psi\frac{\pi}{2} - \phi_s \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\phi_s & -\cos\phi_s \\ 0 & \cos\phi_s & \sin\phi_s \end{bmatrix} \quad \textbf{Equation 41}$$

Equation 42 is the combination of the two rotations using primary rotation order of Z-Y-X, and Equation 43 is the result of combining the two rotations.

$$\underset{R}{^{XYZ\to ENU}}(\phi_s, \lambda_s) = \begin{bmatrix} -\sin(\lambda_s) & -\cos(\lambda_s) & 0 \\ \cos(\lambda_s) & -\sin(\lambda_s) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\phi_s & -\cos\phi_s \\ 0 & \cos\phi_s & \sin\phi_s \end{bmatrix} \quad \textbf{Equation 42}$$

$$\underset{R}{^{XYZ\to ENU}}(\phi_s, \lambda_s) = \begin{bmatrix} -\sin(\lambda_s) & -\cos(\lambda_s)\sin\phi_s & \cos(\lambda_s)\cos\phi_s \\ \cos(\lambda_s) & -\sin(\lambda_s)\sin\phi_s & \sin(\lambda_s)\cos\phi_s \\ 0 & \cos\phi_s & \sin\phi_s \end{bmatrix} \quad \textbf{Equation 43}$$

C.5 Staging System's Rotation Matrices

The Staging system defines its rotational standard by identifying the five factors above:

1) All Staging System rotations use the right-handed system.
2) The Staging System uses Tait-Bryan rotations.
3) The Staging System uses Z-Y-X rotations, i.e., the first rotation is around Z, second rotation is around Y, and third rotation is around X.
4) The Staging System uses Intrinsic rotations, i.e., the first rotation is around Z, the second rotation is around the **new** Y axis, and the third rotation is around the **new** X axis.
5) When computing the root stage's location of a point in any child stage's coordinate system the rotational matrices apply Alibi rotations.

Appendix D Retired Requirements

Requirement	
ST 1906-03 (Retired)	When using an AbsGeocentric instance, and one or more of the x, y, or z attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown.
ST 1906.2-06 (Retired)	When using an AbsGeodetic instance, and one or more of the lat, lon, or hae attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown. *(Sometimes the HAE is unknown when specifying image points)*
ST 1906-04 (Retired)	Unless a document specifies the use of multiple list-terminal stages, a Stage list shall contain at most one (1) list-terminal stage. *(There is a need to have multiple list-terminal stages)*
ST 1906.2-08 (Retired)	When using an AbsEnu instance, and one or more of the rotAboutEast, rotAboutNorth, or rotAboutUp attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown. *(There are devices which may not measure all three axes, e.g., compass, accelerometers)*
ST 1906.2-09 (Retired)	When using an RelOrientation instance, and one or more of the alpha, beta, or psi attributes becomes Unknown (as defined in the Motion Imagery Handbook for Report-On-Change), the receiver shall consider all three attributes as Unknown. *(A stage may not know all three axes at once)*